Autonomic Nervous System

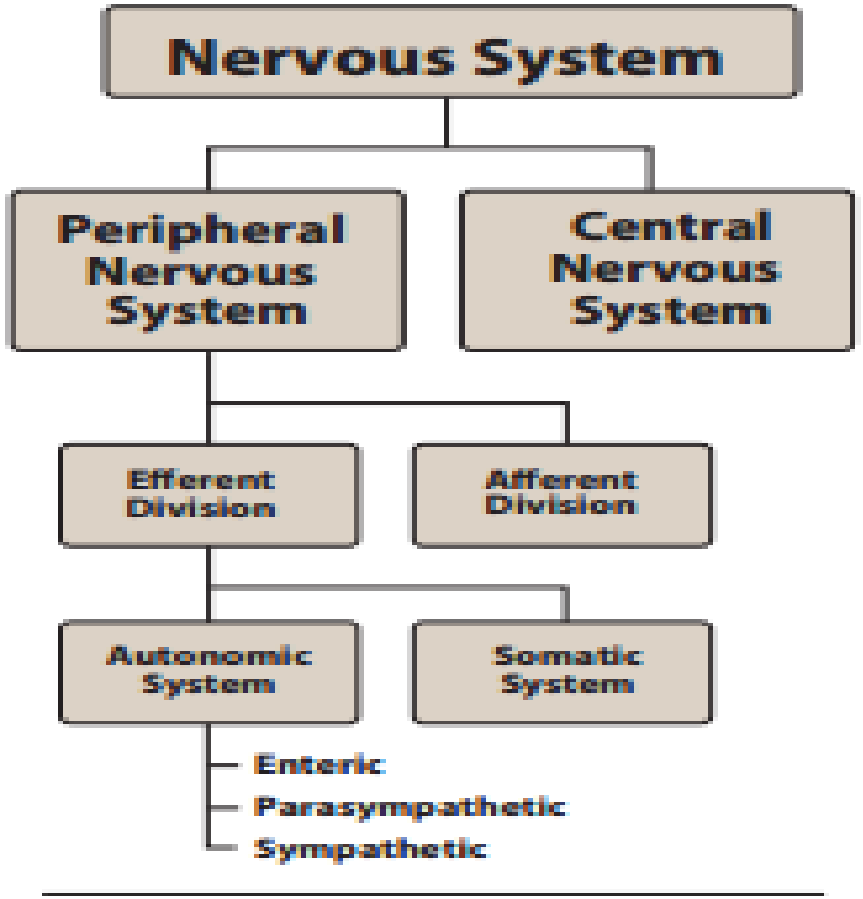
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Learning objectives

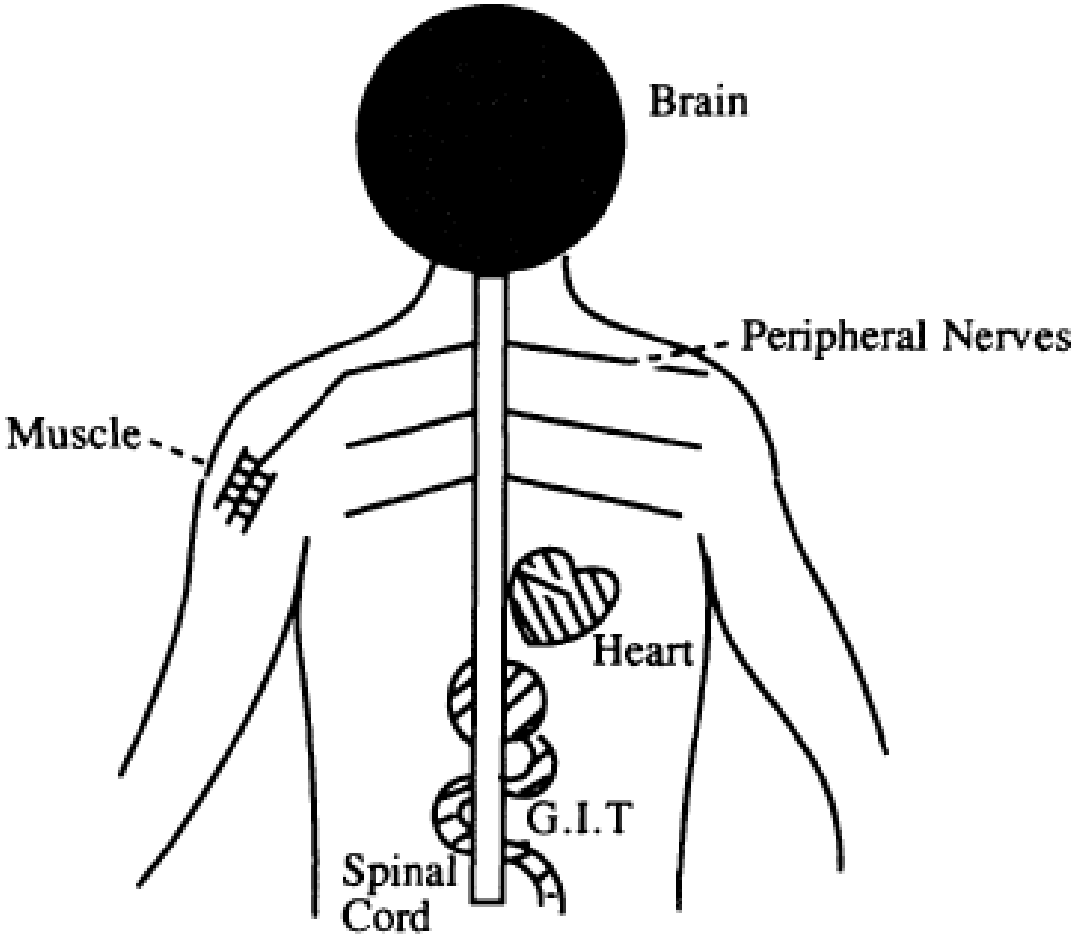
At the end of this session, students are able to:-

- ✓Differentiate the neurotransmitters used in both sympathetic and parasympathetic nervous system
- ✓Discuss the types of cholinergic receptors and their locations
- ✓Explain the SAR of acetylcholine
- ✓Explain why acetylcholine is prone to hydrolysis

The nervous system



Nervous system cont'd…



Nervous system cont'd…

- The motor nervous system is further sub divided into the somatic and autonomic
  - Somatic
    - Innervating the skeletal muscle with no synapses and gap until the skeletal muscle
    - The neurotransmitter released is acetylcholine
    - Acetylcholine binds to cholinergic receptors within the cell membrane of muscle cell
    - The final effect of such innervation is contraction of the skeletal muscle

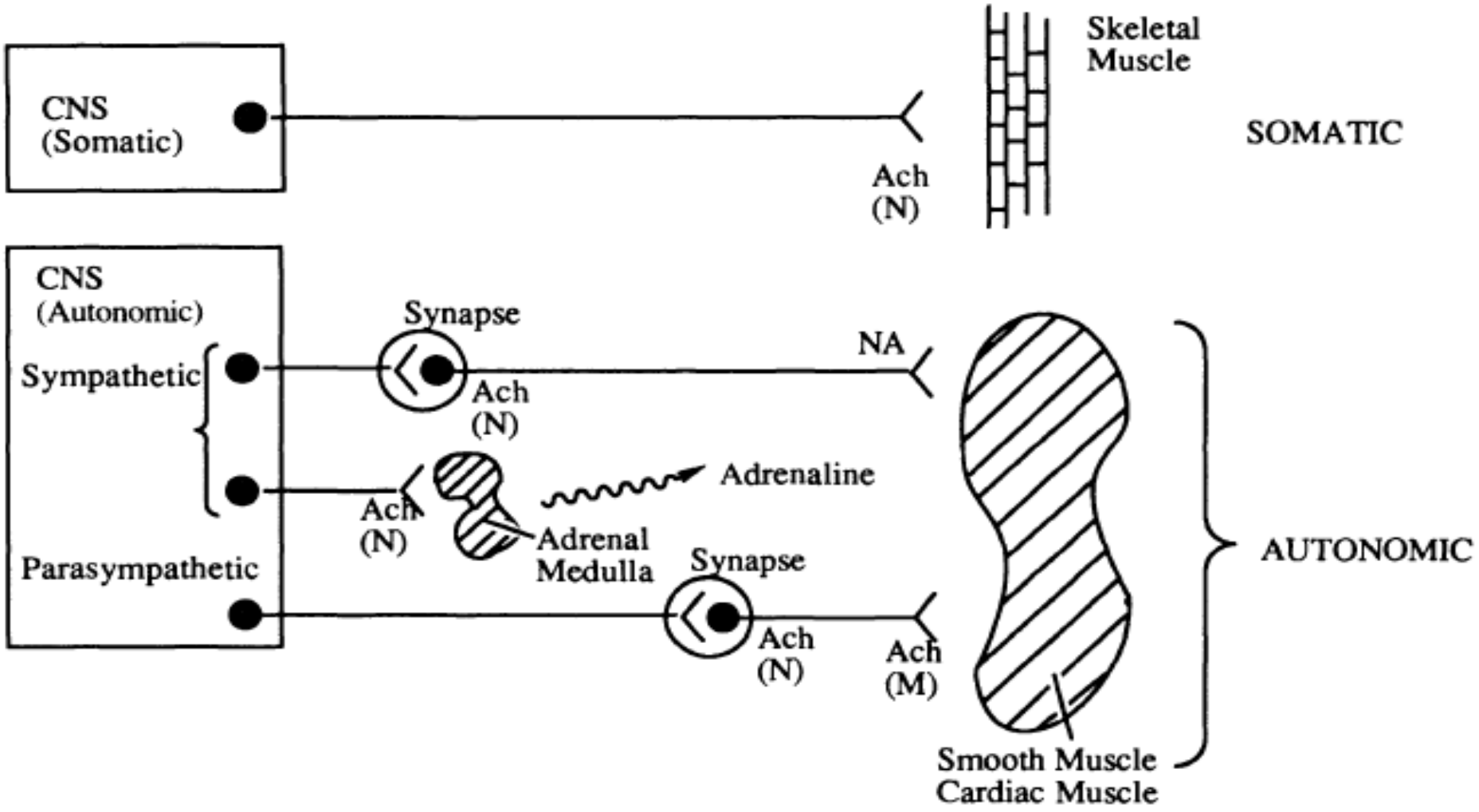


Figure 2: Motor nerves of the peripheral nervous system

Nervous system cont'd…

- The autonomic nervous system mainly innervates
  - smooth muscles,
  - cardiac muscles and
  - the adrenal medulla
- This system can be further subdivided into the
  - Parasympathetic
  - Sympathetic

Nervous system cont'd…

- Parasympathetic

  - The preganglionic fibers are long, and the postganglionic ones are short
  - Travel a long distance and then connect with another nerve that innervates the smooth muscles.
  - The neurotransmitter released here is acetylcholine

Nervous system cont'd…

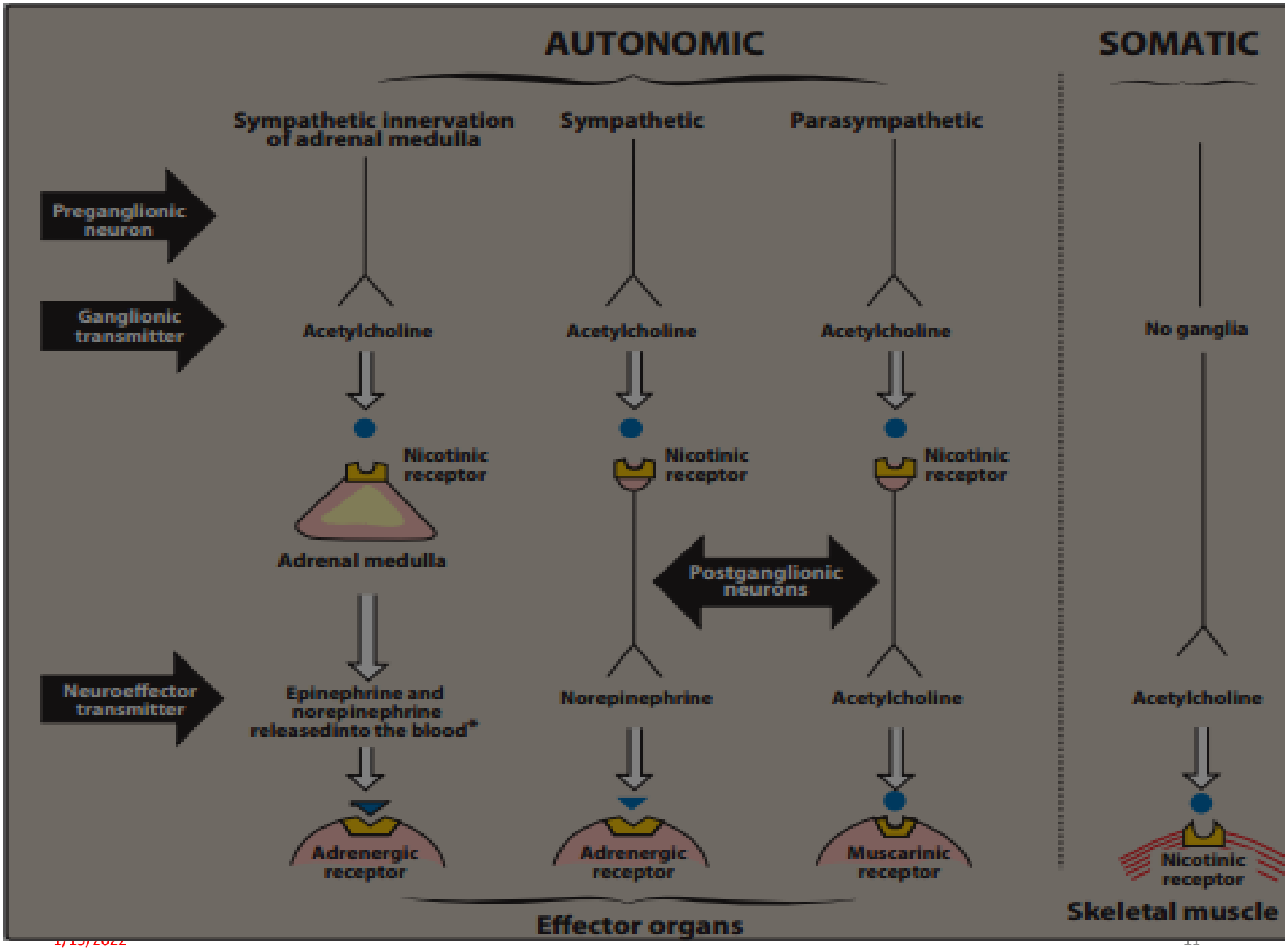
- Sympathetic nervous system

  - The preganglionic fibers are short, and the postganglionic ones are long
  - Almost immediately synapse with another nerve cell.
  - Acetylcholine is the neurotransmitter.
  - Then the second nerve innervates the smooth muscles but releases a different neurotransmitter, Noradrenaline.

Nervous system cont'd…

- The ones that innervate the adrenal medula release noradrenaline but adrenal medula releases the hormone adrenaline

  - Adrenaline circles the body carrying messages to cells that are not innervated.


AUTONOMIC
SOMATIC
Sympathetic innervation of adrenal medulla
Sympathetic
Parasympathetic
Preganglionic neuron
Ganglionic transmitter
Acetylcholine
Acetylcholine
Acetylcholine
No ganglia
Nicotinic receptor
Nicotinic receptor
Nicotinic receptor
Adrenal medulla
Postganglionic neurons
Neuroeffector transmitter
Epinephrine and norepinephrine releasedinto the blood*
Norepinephrine
Acetylcholine
Acetylcholine
Adrenergic receptor
Adrenergic receptor
Muscarinic receptor
Nicotinic receptor
Effector organs
Skeletal muscle

Nervous system cont'd…

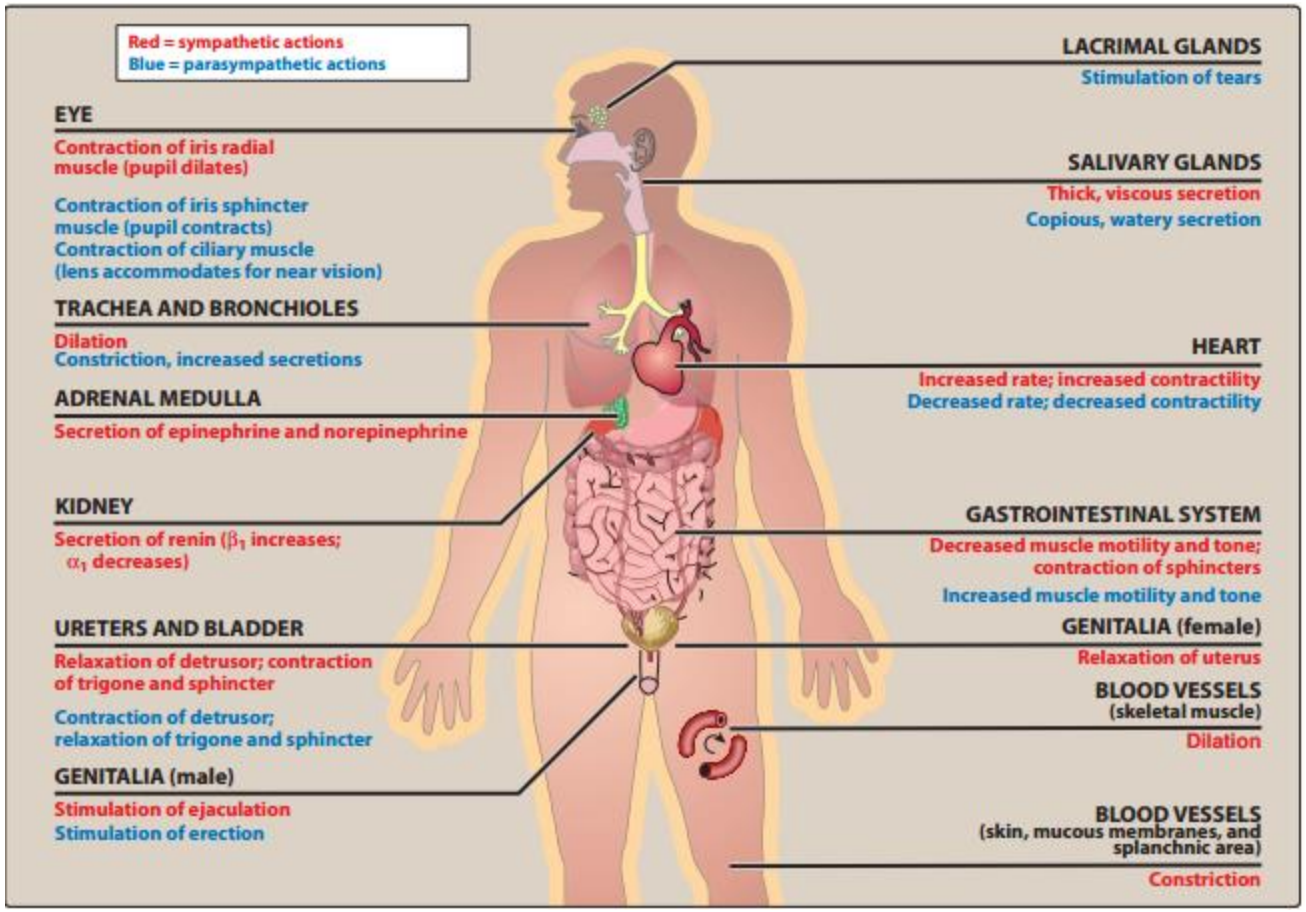
- The enteric system is located in the walls of the intestine

  - It is a collection of nerve fibers that innervate the gastrointestinal (GI) tract, pancreas, and gallbladder, and it constitutes the "brain of the gut."
  - Receives messages from sympathetic and parasympathetic nerves
  - Also responds to local effects to provide local reflex pathways which are important in the control of GIT function.
  - Examples are Serotonin, neuropeptides and ATP

Actions of the peripheral nervous system

- Somatic
  - Innervates the skeletal muscle
- Autonomic
  - Parasympathetic, one nerve and acetylcholine released, “rest” response
  - Sympathetic, first one nerve with acetylcholine released then another nerve with noradrenaline released
    - Is a “fight or flight” response. Digestion is reduced, heart rate is increased

Cont…


Red = sympathetic actions
Blue = parasympathetic actions
EYE
Contraction of iris radial muscle (pupil dilates)
Contraction of iris sphincter muscle (pupil contracts)
Contraction of ciliary muscle (lens accommodates for near vision)
TRACHEA AND BRONCHIOLES
Dilation
Constriction, increased secretions
ADRENAL MEDULLA
Secretion of epinephrine and norepinephrine
KIDNEY
Secretion of renin (β_1 increases; α_1 decreases)
URETERS AND BLADDER
Relaxation of detrusor; contraction of trigone and sphincter
Contraction of detrusor; relaxation of trigone and sphincter
GENITALIA (male)
Stimulation of ejaculation
Stimulation of erection
LACRIMAL GLANDS
Stimulation of tears
SALIVARY GLANDS
Thick, viscous secretion
Copious, watery secretion
HEART
Increased rate; increased contractility
Decreased rate; decreased contractility
GASTROINTESTINAL SYSTEM
Decreased muscle motility and tone; contraction of sphincters
Increased muscle motility and tone
GENITALIA (female)
Relaxation of uterus
BLOOD VESSELS (skeletal muscle)
Dilation
BLOOD VESSELS (skin, mucous membranes, and splanchnic area)
Constriction

Peripheral cont'd…

- Para sympathetic and sympathetic nerves are a system of check and balance
  - One regulates the other.
  - If the effect of one is larger than the other then which ever prevails takes the upper hand

Peripheral cont'd…

- Application of an agonist can yield the effect desired, theoretically
  - However, because these nerves and receptors are available all over the body there is a problem of selectively addressing a specific tissue.

Neurotransmitters

- Neurotransmitters are chemicals

  - Which facilitate the communication between nerve cells, and between nerve cells and effector organs.
  - They carry out a crucial role in nerve transmission.
  - Allows the medicinal chemist to design and synthesize organic compounds which can mimic (agonists) or block (antagonists) the natural neurotransmitters.

Types of neurotransmitters

- Over fifty signal molecules in the nervous system have tentatively been identified

- six signal compounds, including norepinephrine (and the closely related epinephrine), acetylcholine, dopamine, serotonin, histamine, and γ-aminobutyric acid(GABA), are most commonly involved in the actions of therapeutically useful drugs.

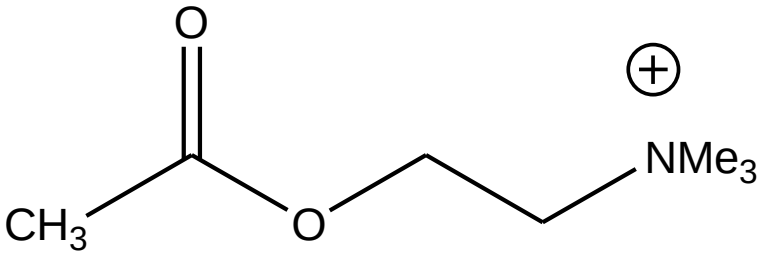
Cont'd

- Acetylcholine and norepinephrine are the primary chemical signals in the ANS.

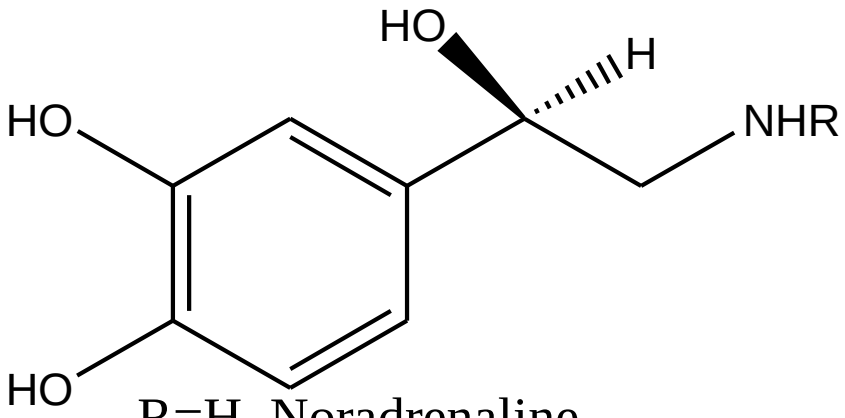
- Not only are these neurotransmitters released on nerve stimulation, but also cotransmitters, such as adenosine, often accompany them and modulate the transmission process.

Neurotransmitters

- Only two will be considered,
  - Acetylcholine and
  - Noradrenaline
  - Adrenaline



Acetylcholine



R=H, Noradrenaline
R=Me, Adrenaline

The cholinergic nervous system

The cholinergic system

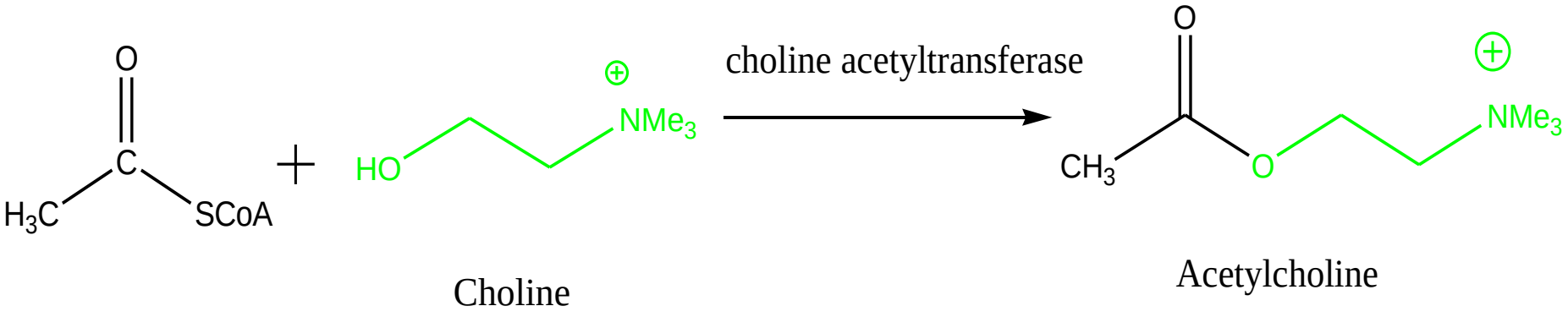
- Neurotransmission at cholinergic neurons involves six sequential steps

1. Biosynthesis of acetylcholine (from choline and acetyl coenzyme A)

- Choline is transported from the extracellular fluid into the cytoplasm of the cholinergic neuron by an energy-dependent carrier system that cotransports sodium
  - ✓Choline has a quaternary nitrogen and carries a permanent positive charge, and, thus, can not diffuse through the membrane
  - ✓The uptake of choline is the rate-limiting step in ACh synthesis

The cholinergic system

- Acetyl CoA is derived from the mitochondria and is produced by the pyruvate oxidation and fatty acid oxidation.



Choline

Acetylcholine

Cholinergic cont'd…

2. Incorporation into membrane bound vesicles by means of a specific carrier protein
3. Release of the transmitter into the synaptic gap

- ✓ When an action potential arrives at a nerve ending, voltage-sensitive calcium channels on the presynaptic membrane open, causing an increase in the concentration of intracellular calcium.
- ✓ Elevated calcium levels promote the fusion of synaptic vesicles with the cell membrane and the release of their contents into the synaptic space

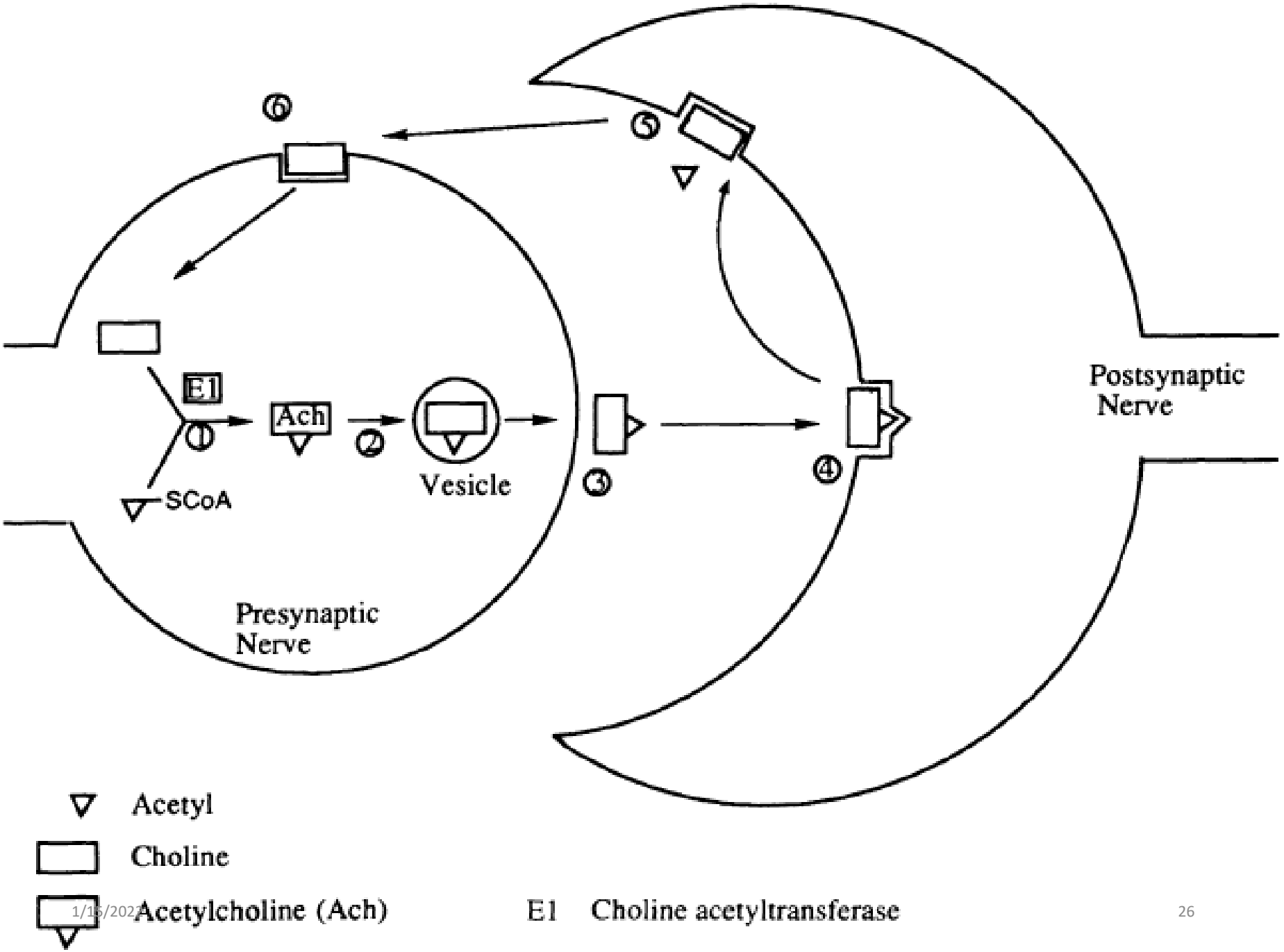
Cholinergic system

4. Binding to the cholinergic receptor and stimulation of the second nerve

5. Hydrolysis by acetylcholinesterase to choline and acetate.

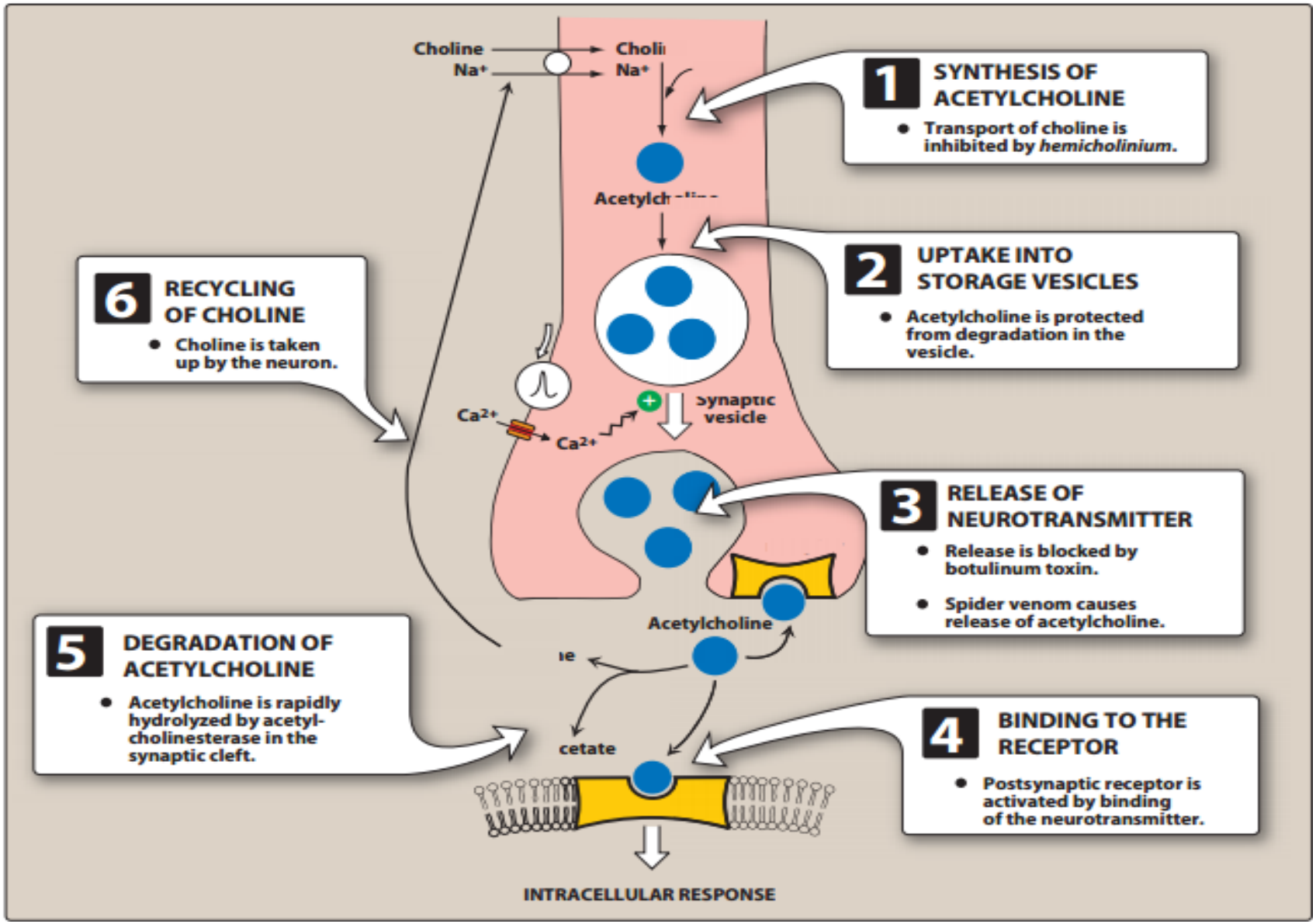
- ✓ [Note: Butyrylcholinesterase, sometimes called pseudocholinesterase, is found in the plasma but does not play a significant role in the termination of ACh's effect in the synapse.]

6. Choline taken up into the presynaptic nerve, cycle continues


⑥
⑤
E1
①
Ach
②
Vesicle
③
④
SCoA
Presynaptic Nerve
Postsynaptic Nerve
Acetyl
Choline
Acetylcholine (Ach)
E1 Choline acetyltransferase

Cholinergic cont'd…

- Therefore, there are several stages that can be targeted to promote/inhibit the overall process
  - So far stages 4 and 5 are the most widely used and analyzed


Choline
Na+
Choli
Na+
1 SYNTHESIS OF ACETYLCHOLINE
Transport of choline is inhibited by hemicholinium.
Acetylch
2 UPTAKE INTO STORAGE VESICLES
Acetylcholine is protected from degradation in the vesicle.
6 RECYCLING OF CHOLINE
Choline is taken up by the neuron.
synaptic vesicle
Ca2+
Ca2+
3 RELEASE OF NEUROTRANSMITTER
Release is blocked by botulinum toxin.
Spider venom causes release of acetylcholine.
Acetylcholine
5 DEGRADATION OF ACETYLCHOLINE
Acetylcholine is rapidly hydrolyzed by acetyl-cholinesterase in the synaptic cleft.
cetate
4 BINDING TO THE RECEPTOR
Postsynaptic receptor is activated by binding of the neurotransmitter.
INTRACELLULAR RESPONSE

Cholinergic receptors

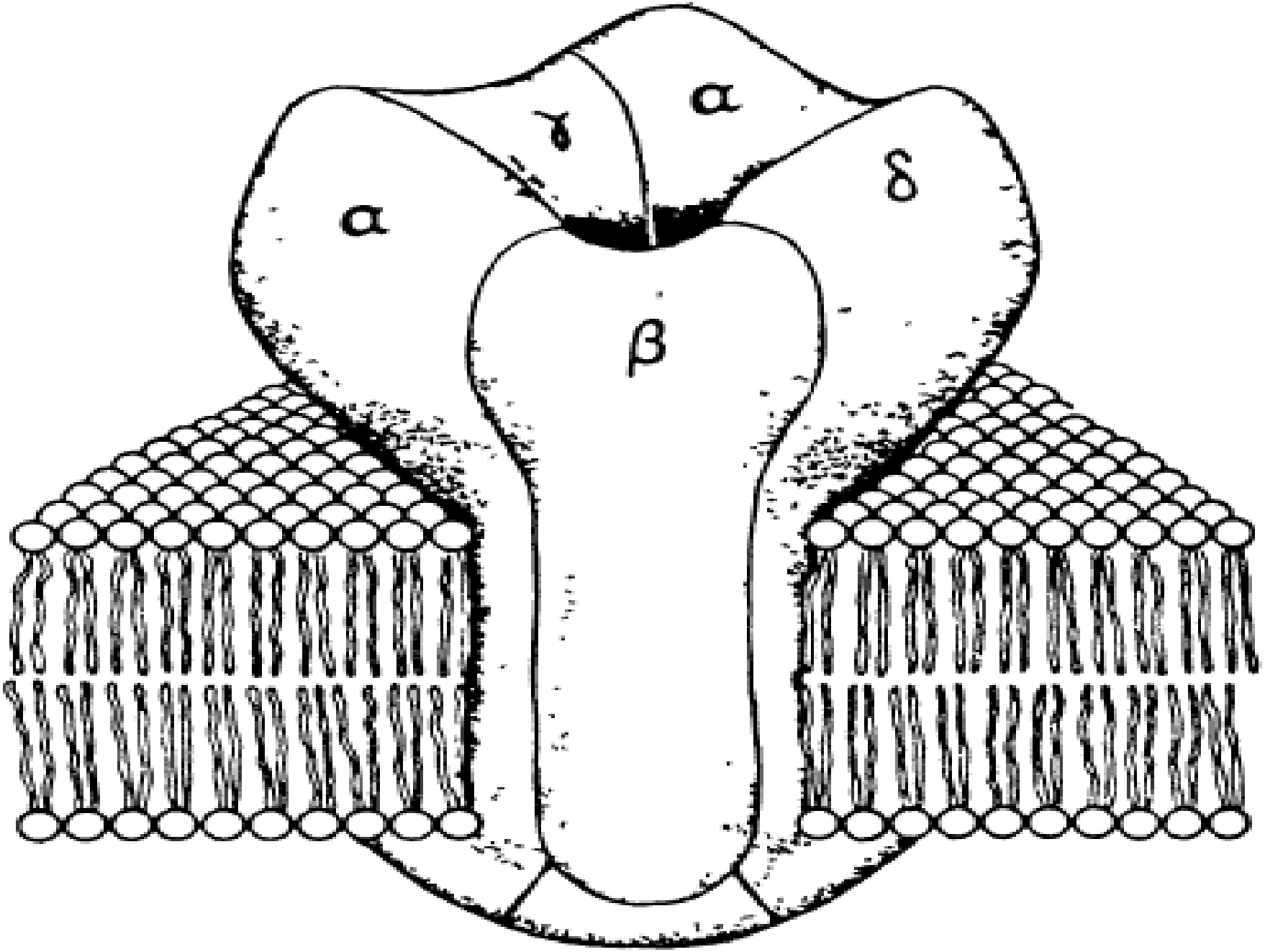
- Natural compounds screening indicated that there are different types of the cholinergic receptor

- Extracts from tobacco, nicotine, and poisonous mushrooms, muscarine were both agonists of the acetylcholine receptor but that they had different physiological response

1. Nicotinic receptors
2. Muscarinic receptors

The nicotinic receptor structure

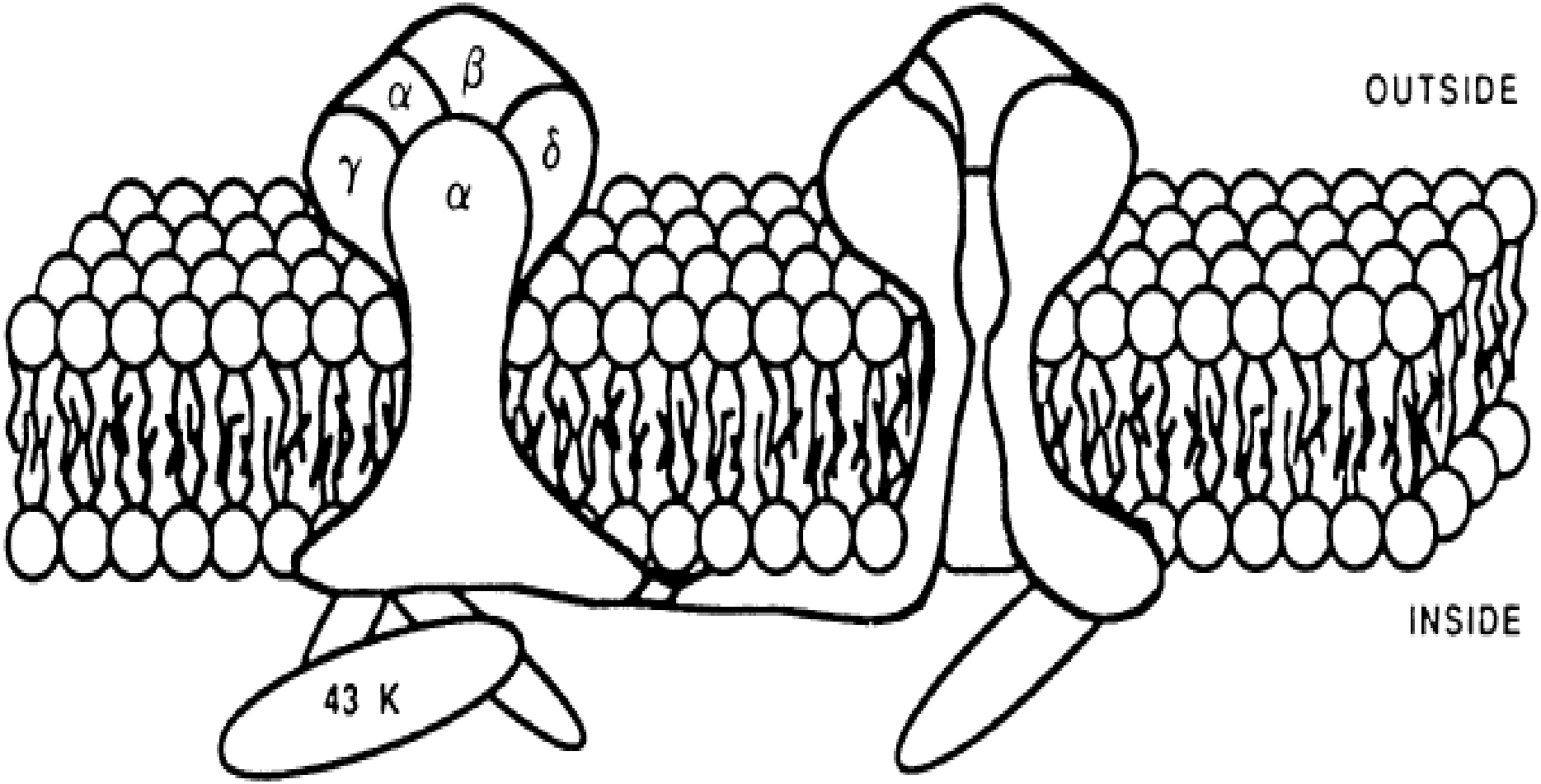
- It is a protein subunit made up of five subunits, two of which are the same
- They are:-
  - Two alpha
  - One beta
  - One gamma
  - One delta
- It forms a cylindrical or barrel shape which traverses the cell membrane


γ
α
α
δ
β

Cont…

- The center serves as an ion channel for sodium
  - A lock system is controlled by the interaction of the receptor with acetylcholine
  - When acetylcholine is unbound the gate is shut and vice versa
- Nicotinic receptors occur in pairs linked together by a disulfide bridge between the delta subunits
- The binding site for acetylcholine is situated on the alpha subunit
  - So how many binding sites per receptor proteins?

Nicotinic receptor structure



The muscarinic receptor-structure

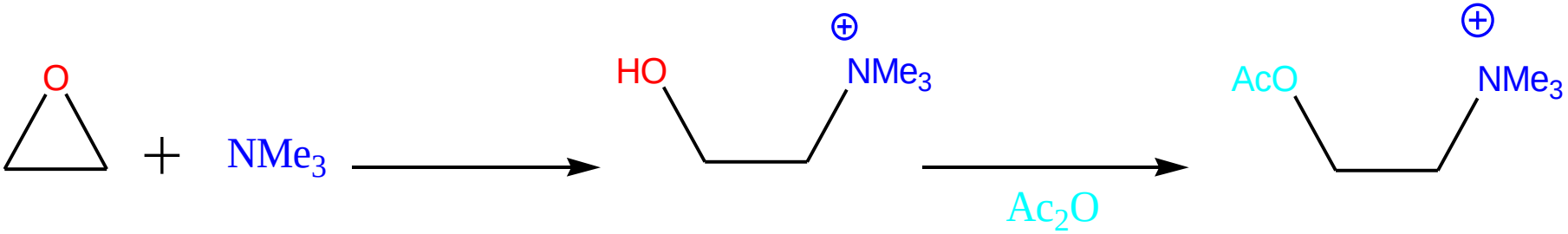
- Is more difficult than nicotinic receptors to isolate
- Muscarinic receptors have been commonly subdivided into three subgroups
  - M1
  - M2 and
  - M3
  - M4?
  - M5?

Cont…

- M2 receptors are located in heart muscle and parts of the brain
  - Acts by controlling the enzyme synthesis of secondary messengers rather than by directly controlling an ion channel
- Muscarine was active at smooth muscle and cardiac muscle

Agonists at the cholinergic receptor

- If there is a lack of acetylcholine, why don't we just administer more acetylcholine considering that it is easy to synthesize?

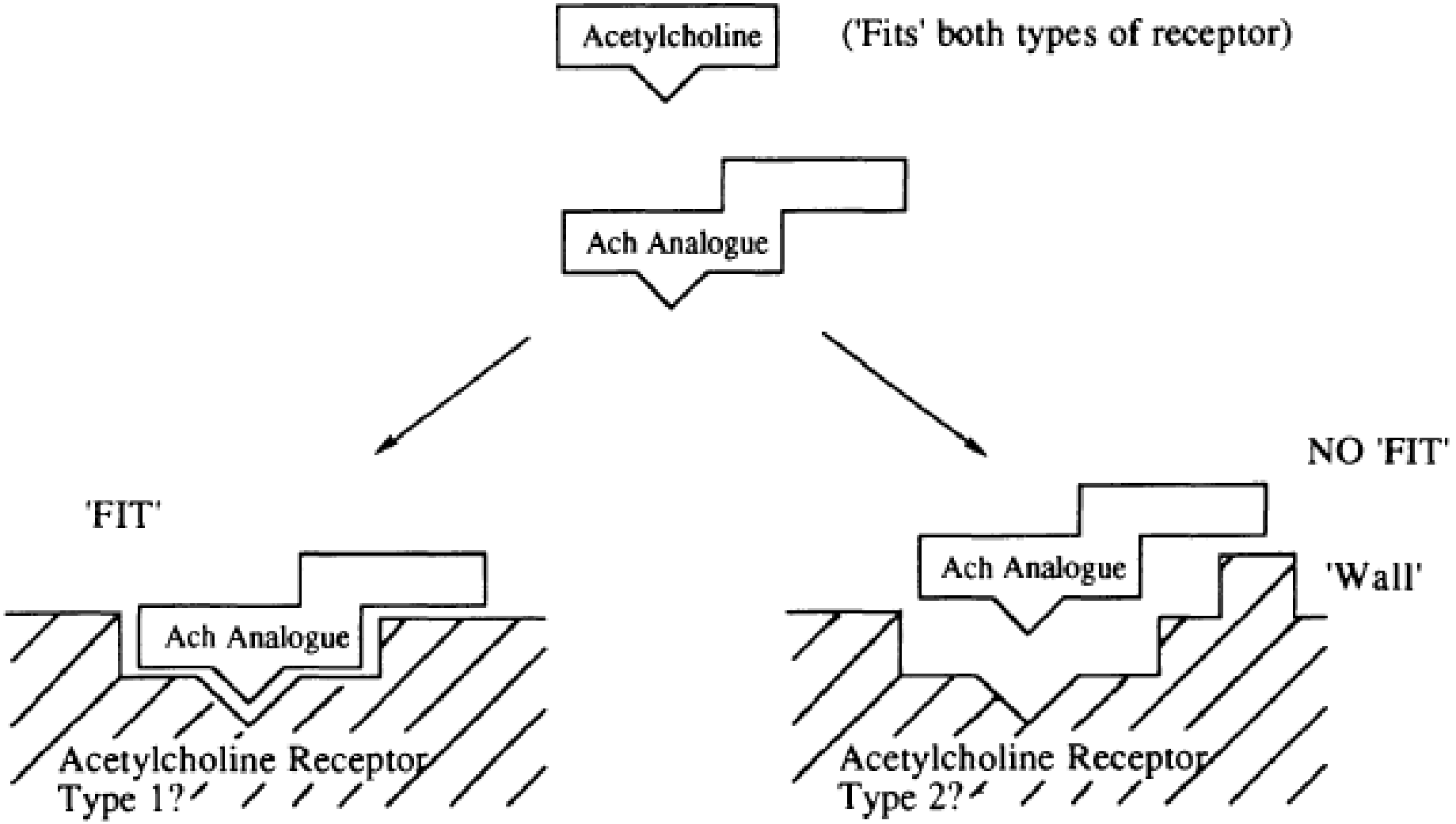


Agonists cont'd…

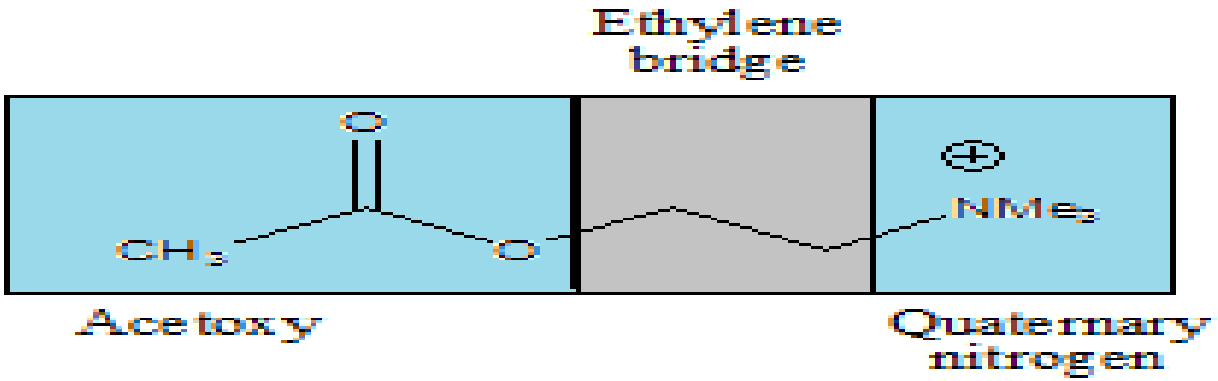
- There are three reasons why this is not feasible,
  - Easily hydrolyzed in the stomach by acid catalysis,
    - Not suitable for oral administration
  - Acetylcholine is hydrolyzed in the blood easily chemically or by enzyme (esterase)
    - Not suitable for IV/IM either
  - Non selectivity
    - Because the receptor is present all over the body, acetylcholine can switch on all receptors at multiple sites

Agonists cont'd…

- An analogue that is both selective and stable to hydrolysis is required
- Selectivity
  - Site distribution may be different
    - E.g. Adipose tissues tend to accumulate lipophilic drugs
    - Hydrophilic drugs have hard time crossing the BBB
  - Receptors might be slightly different in various parts of the cell

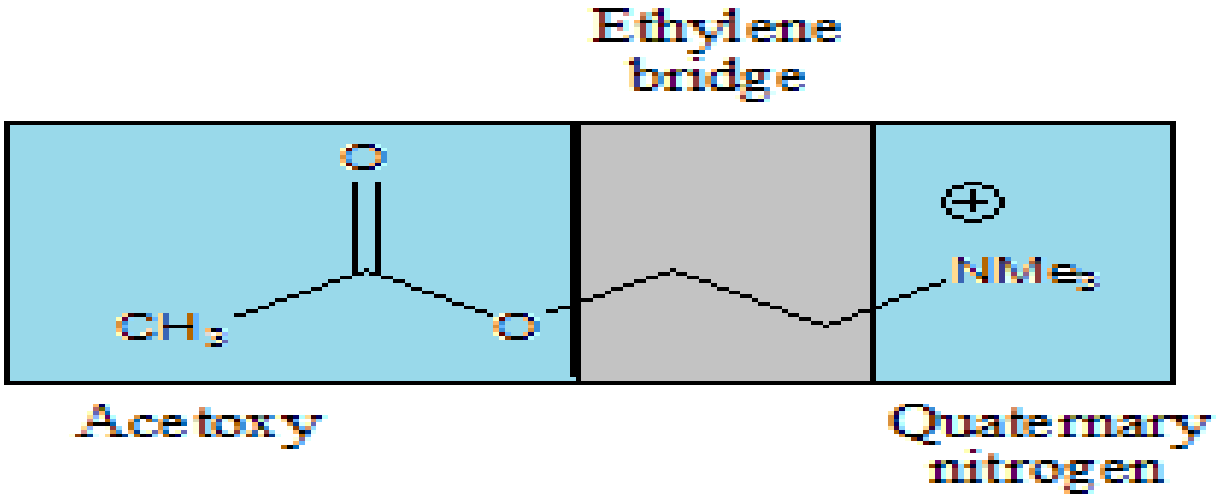

Acetylcholine
('Fits' both types of receptor)
Ach Analogue
'FIT'
Ach Analogue
Acetylcholine Receptor
Type 1?
NO 'FIT'
Ach Analogue
'Wall'
Acetylcholine Receptor
Type 2?

SAR and receptor binding



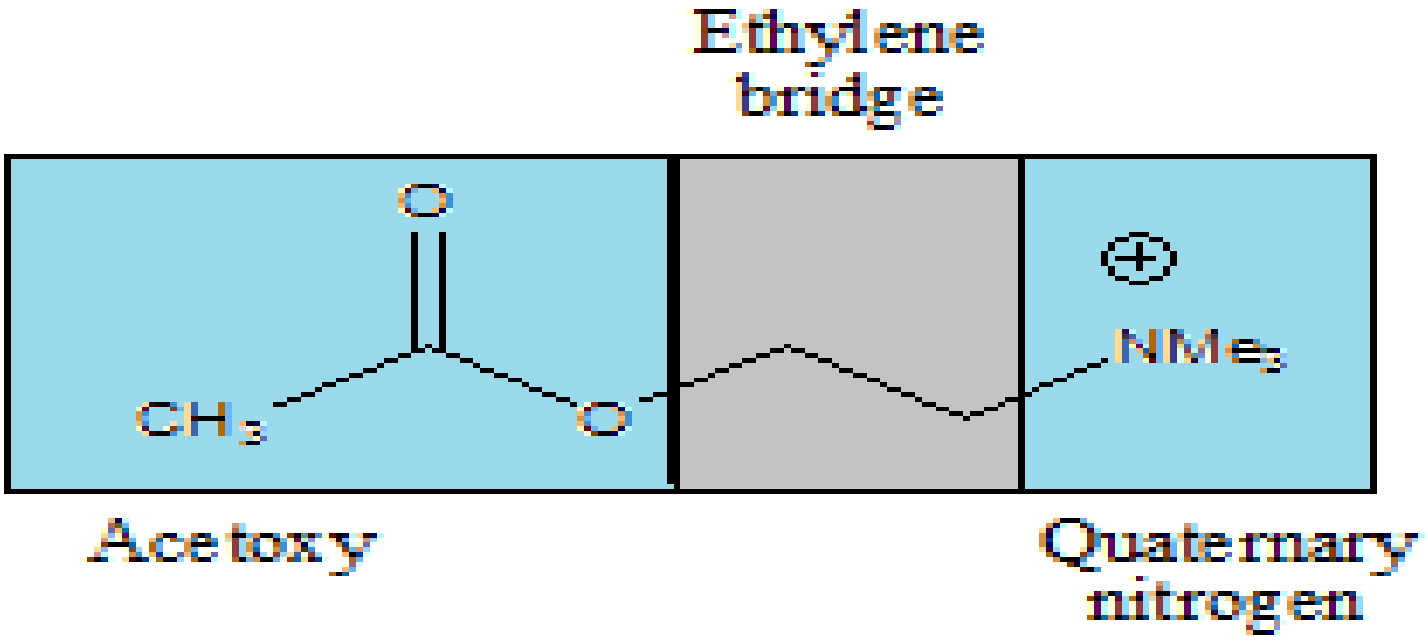
- The positively charged nitrogen atom is essential to activity
  - ✓Replacing it with a neutral carbon atom eliminate the activity
- The distance from the nitrogen to the ester group is important

SAR



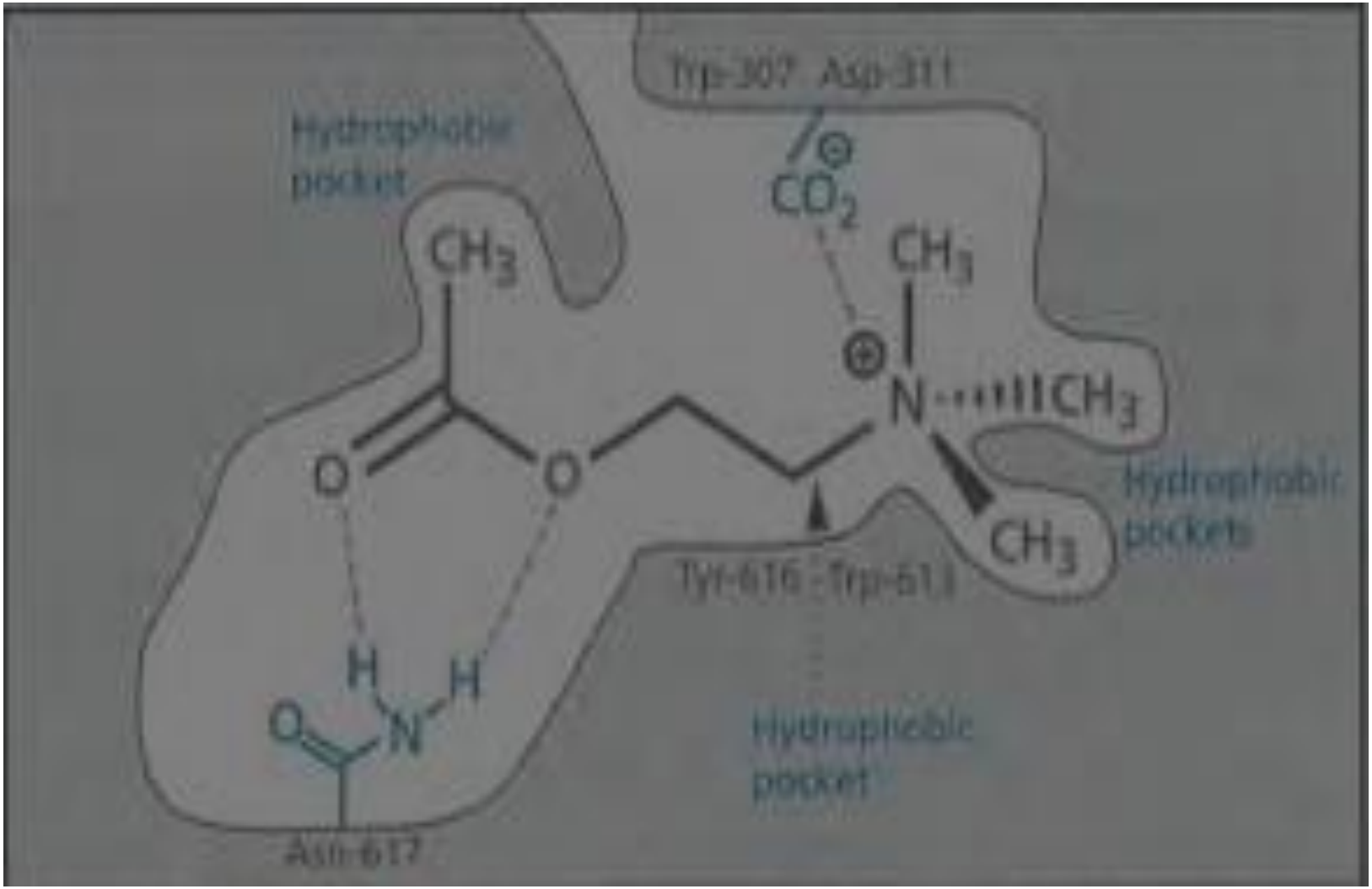
- The ester functional group is important
- The overall size of the molecule can not be altered greatly.
  - Bigger molecules have poorer activity
- The ethylene bridge between the ester and the nitrogen atom cannot be extended

SAR



- There must be two methyl groups on the nitrogen
  - ✓A larger, third alkyl group is tolerated, but more than one large alkyl group leads to loss of activity
- Bigger ester groups lead to a loss of activity

Muscarinic receptor binding site



Conclusions

- Hydrogen bonding between the ester group of acetylcholine and an asparagine residue
- Hydrophobic pocket accommodating the methyl group of the ester, but nothing larger.
  - ✓Thought to be more important in muscarinic receptors than nicotinic receptors
- A strong ionic interaction between the charged nitrogen atom and the anionic side group of the aspartic acid residue.
- There may also be an induced dipole interaction between the NMe_3^+ group and the aromatic residues in the hydrophobic pocket

Conclusion cont'd…

- How is an induced dipole interaction between a charged moiety and another groups possible?
  - The positive charge is not localized on the nitrogen atom
    - Distributed through the three methyl groups
  - The aromatic rings of the receptor can be stabilized by binding to NMe_3^+ .
    - The electrons on the aromatic rings are diffused into the NMe_3^+ group, inducing a dipole moment that account for the binding

Cont…

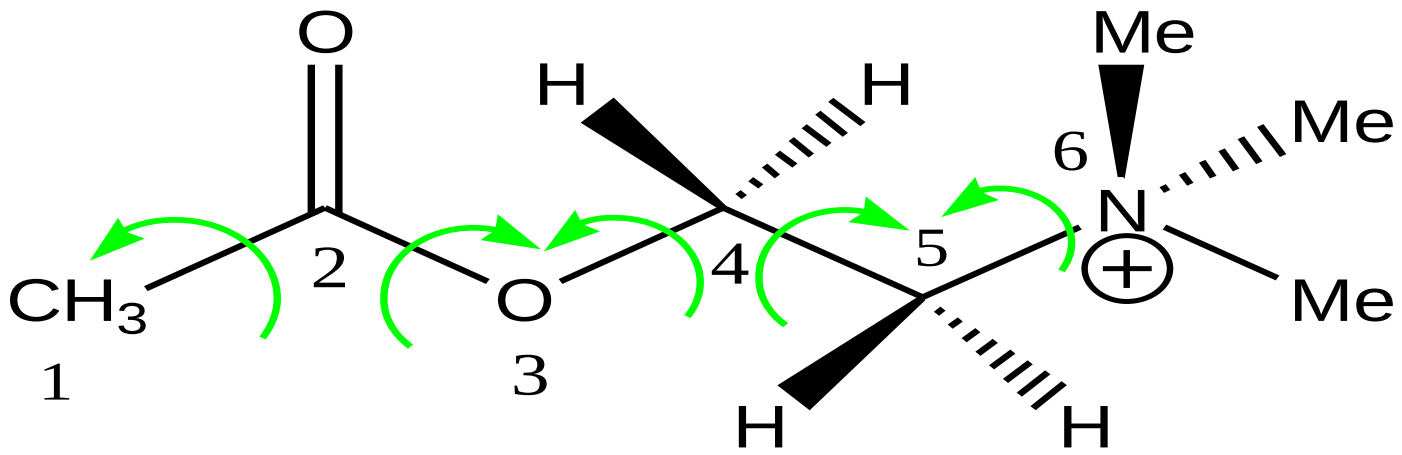
- The NMe_3^+ group is placed in hydrophobic pocket lined with three aromatic amino acid
  - ✓The pocket also contain two hydrophobic pockets which are large enough to accommodate two of the three methyl substituents on the NMe_3^+
  - ✓The third methyl substituents on nitrogen positioned in an open region of binding site
    - Its possible to replace it with other group

At the end of this session, students are able to:-

- ✓Discuss the role of rigidity
- ✓Explain why acetylcholine is prone to hydrolysis
- ✓Discuss the role of steric shield and electronic effect in producing stable analogues of acetylcholine
- ✓List the clinical use of muscarinic and nicotinic agonists
- ✓Explain the effect of tertiary nitrogen on the activity of pilocarpine
- ✓List the clinical use of both muscarinic and nicotinic antagonists
- ✓Explain the SAR of atropine
- ✓Discuss why atropine can cross BBB and how its inhibited

Rigidity

- Acetylcholine is a highly flexible molecule
  - In order to establish the active conformation of acetylcholine, rigid analogs were synthesized

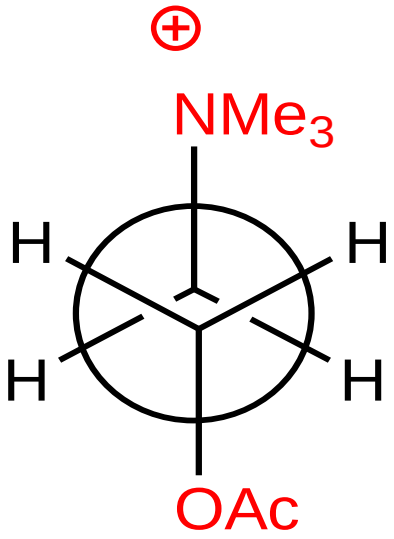


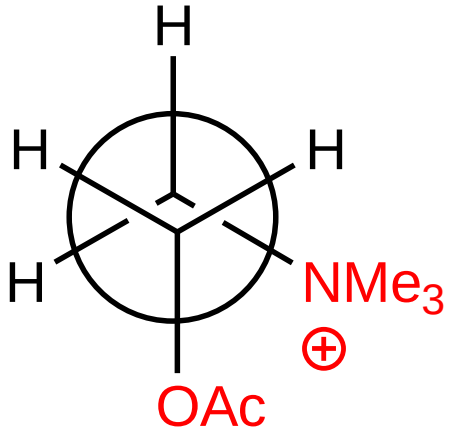
bond rotations in acetylcholine

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Rigidity cont'd…

- In the past, its assumed that a flexible compound such as acetylcholine would adopt its most stable conformation when binding



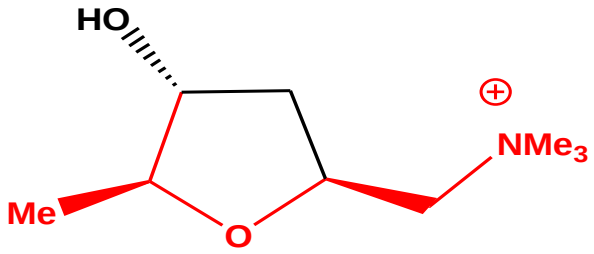


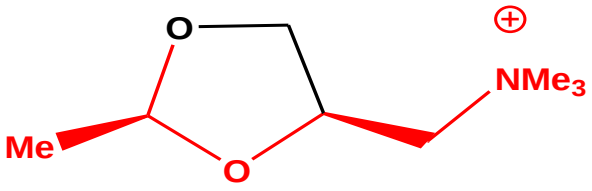
gauche conformation

No significant energy difference

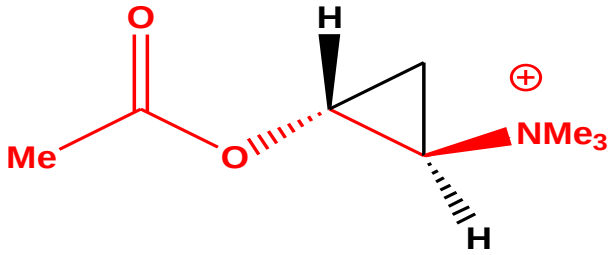
Rigidity cont'd…

- Rigid cyclic molecules were designed containing the skeleton of acetylcholine to establish the active conformation of acetylcholine
  - ✓Muscarine and some of its analogs are shown below





Muscarine

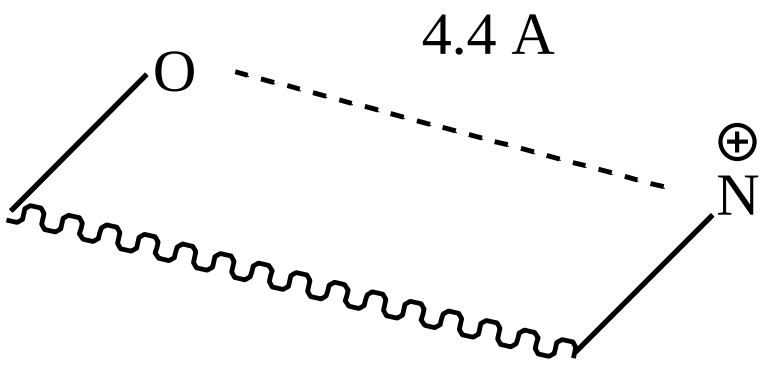


Rigidity cont'd…

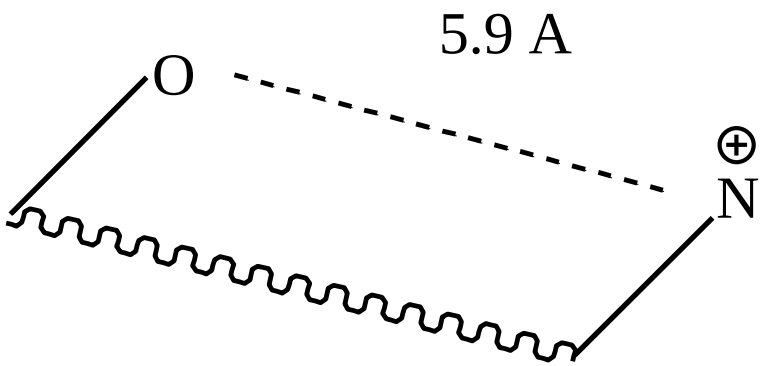
- Many such structures have been prepared but it has not been possible to identify one specific active conformation for acetylcholine.
  - This indicates that the cholinergic receptor has a certain amount of latitude for acetylcholine.
  - The distance between the ester group and the quaternary nitrogen is important for binding and differ for muscarinic and nicotinic receptor.

Rigidity cont'd…

- For muscarinic and nicotonic receptors the distance between the ketonic oxygen and the quaternary nitrogen should be 4.4 A^o and 5.9 A^o , respectively



Muscarinic receptor

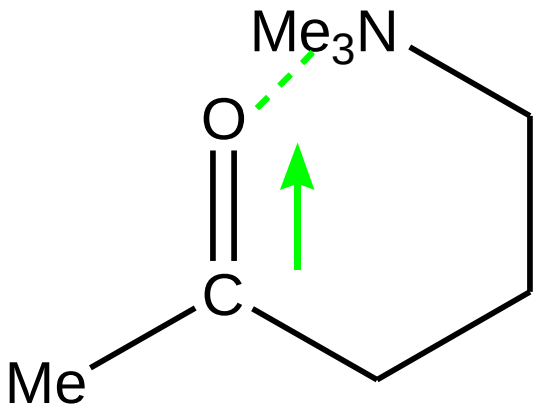


Nicotinic receptor

Pharmacophore of acetylcholine

The instability of acetylcholine

- Acetylcholine is prone to hydrolysis .
  - This is mainly because of the electron withdrawing effect of nitrogen and its interaction with carbonyl group ⊕



neighbouring group paticipation

Instability cont'd…

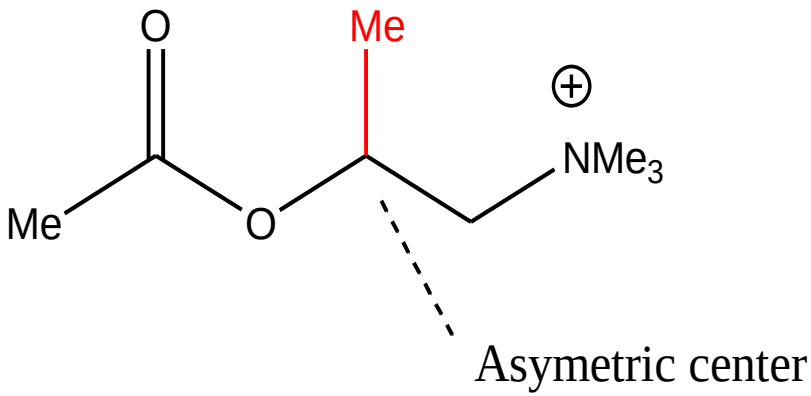
- Nitrogen pulls electrons from oxygen. The oxygen in turn pulls electrons from the carbonyl group of carbon making it even more electron deficient.
  - This carbon is prone to nucleophilic attack.
    - ✓Water is a poor nucleophile but would hydrolyze acetylcholine
  - The influence of nitrogen is known as neighboring group participation or anchimeric assistance

Design of acetylcholine analogs

- To prevent the enzyme or water molecules from attacking acetylcholine two possible approach can be used
  - steric shield techniques
  - Electronic stabilization techniques

Steric shield

- Demonstrated by methacholine
  - A methyl group was attached to the ethylene bridge, this hinders the approach of any potential nucleophile and also hinders binding to esterase enzymes



Methacholine

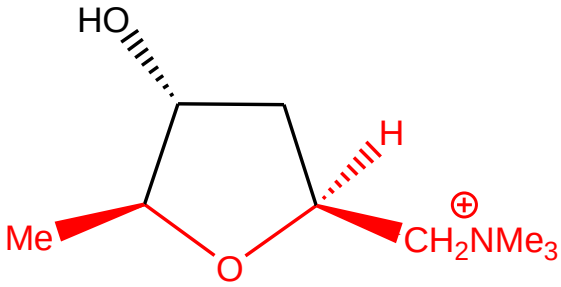
Steric shield

- Methacholine is three times more stable than acetylcholine

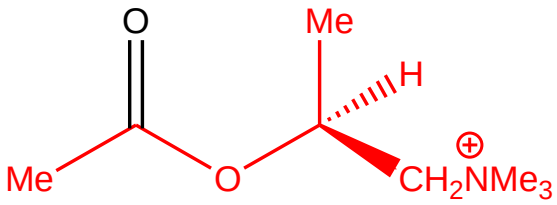
- Many other analogues were tried with bigger and bulkier substituent groups but they had lower cholinergic activity but stability increased.
- Why?

Steric shield

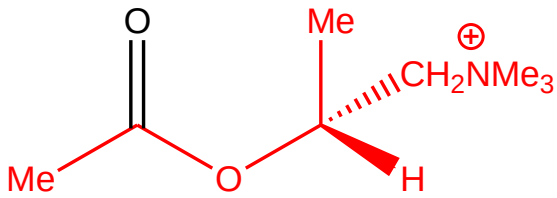
- The introduction of the methyl group also resulted in selectivity towards muscarinic receptor as opposed to nicotinic receptor



Muscarine



(S)-Methacholine



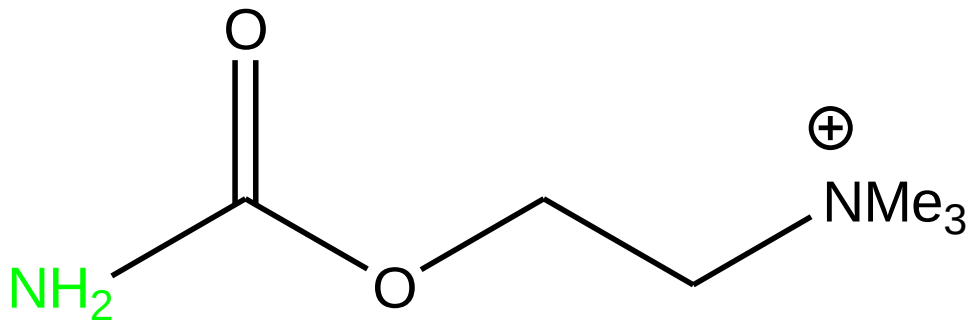
(R)-Methacholine

comparison of muscarine and methacholine

- The S enantiomer is the most active, resemblance

Electronic effects

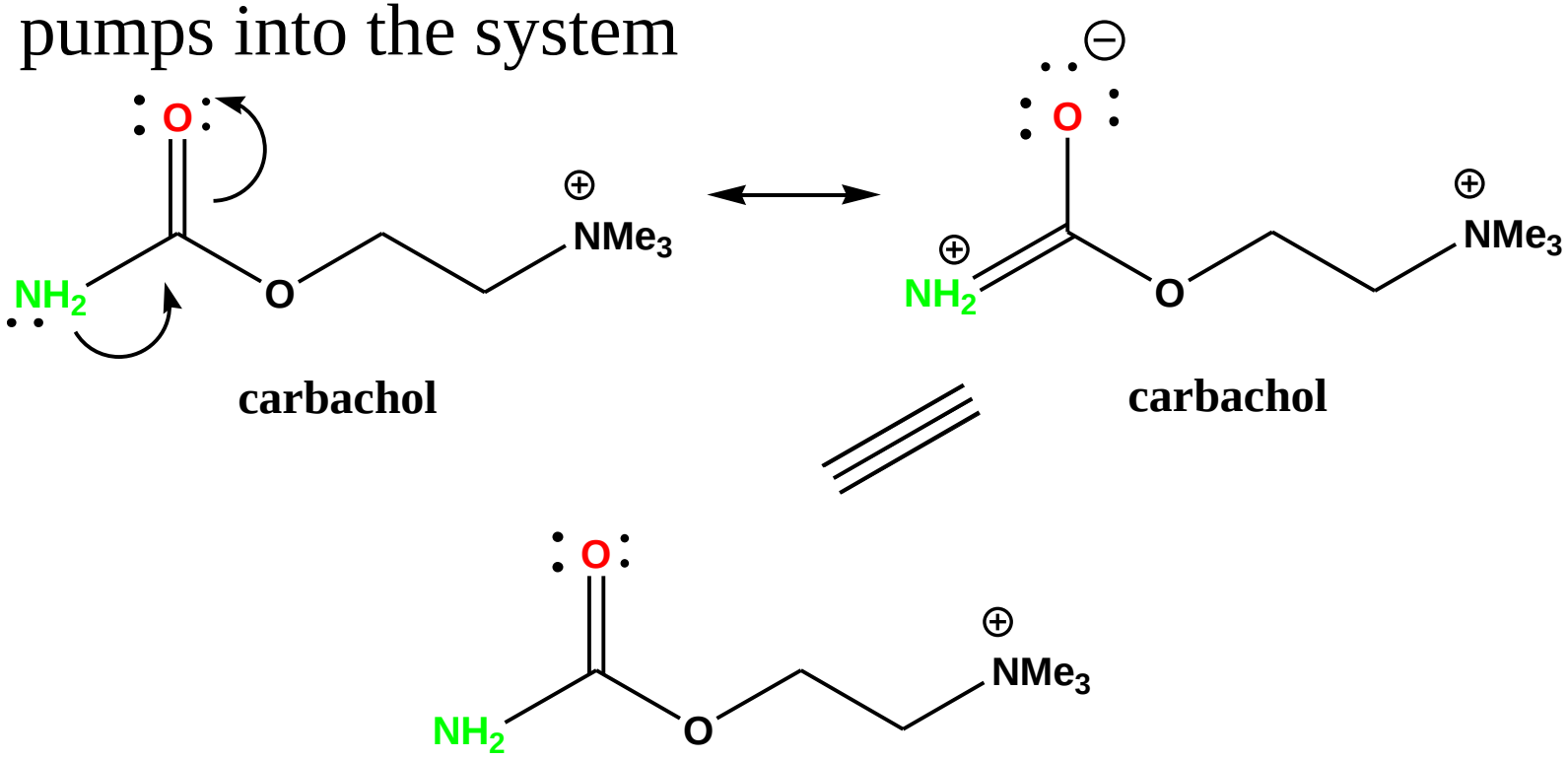
- Carbachol is a long lasting cholinergic agent which is resistant to hydrolysis
  - The acyl methyl group is replaced with NH_2, the ester is replaced by a carbamate or urethane group



carbachol

Electronic cont'd…

- Carbachol is more resistant to hydrolysis because of the lone pair of electrons on the amine group pumps into the system



Electronic cont'd…

- The NH_2 replacement lowers the electrophilicity character of the carbonyl group
  - Carbachol is less selective b/n muscarinic & nicotinic receptors
  - However, its non selectivity is not a problem because it is usually administered locally.

    ✓Carbachol is used in the treatment of glaucoma and it is applied locally to the eye

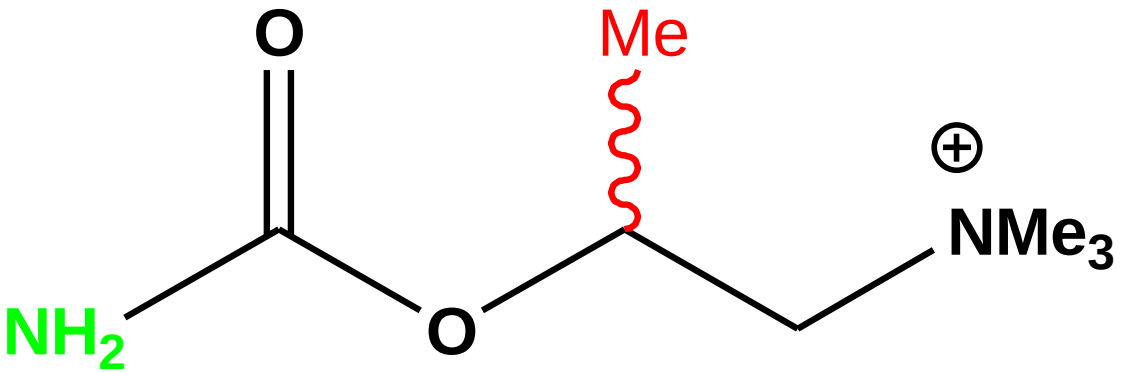
Medicinal che-mystery?

- ❑The enzyme has a hydrophobic pocket site, how does carbachol's hydrophilic amine group bind to the receptor then?

- ✓Bioisosteric replacement of the acyl methyl group with NH2 group. They are both relatively of the same size and they have the same biological effect

Steric shield and electronic stabilization

- Bethanicol is a drug designed by combining the steric shield ability of methanicol and the electronic stabilization technique of carbachol



Bethanicol

Steric and electronic cont'd…

- Bethanicol is both stable to hydrolysis and selective in its action
  - It is occasionally used therapeutically in stimulating the GIT and urinary bladder after surgery
    - ✓Both these organs are shutdown by drug prior to surgery

Clinical use for cholinergic agonists

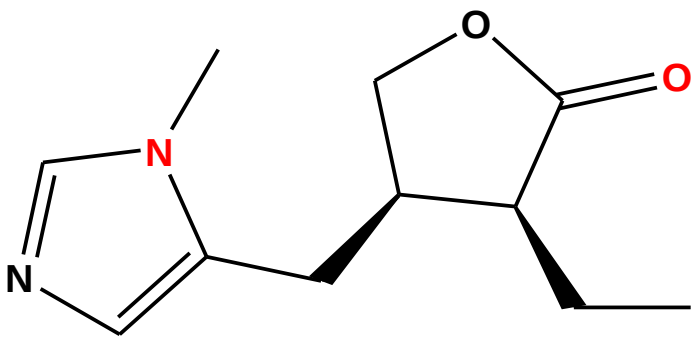
Muscarinic agonists

- The clinical uses for muscarinic agonists are
  - Treatment of glaucoma
  - “Switching on” the GIT and Urinary tract after surgery
  - Treatment of heart defects by decreasing heart muscle activity

Muscarinic analogues

Pilocarpine

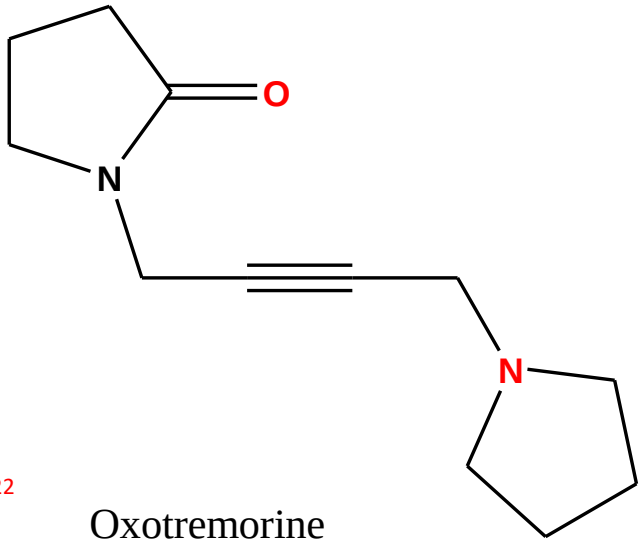
- Used in the treatment of glaucoma
- Although there is no quaternary ammonium group present in pilocarpine, its assumed that the drug is protonated before it interacts with the muscarinic receptor
- Has a pharmacophore of 4.4 A^{o}



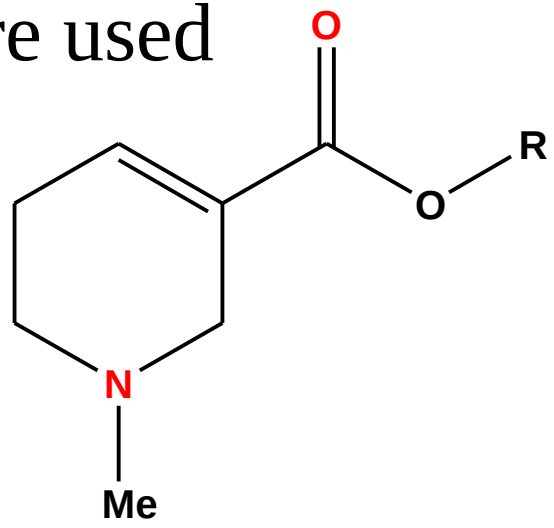
pilocarpine

Analogues cont'd…

- Pilocarpine also considered for Alzheimer's disease
  - Old-timers!!
  - Oxotremorine and various Arecoline analogues are used
- Currently, anticholinesterases are used



Oxotremorine



Arecoline (R=Me) and analogues

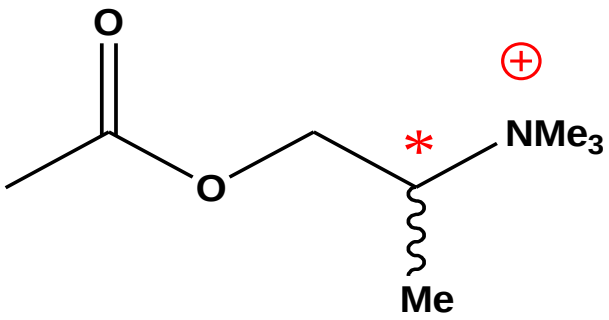
Analogues cont'd…

Nicotinic agonists

- Used in the treatment of myasthenia gravis
  - ❖ an auto immune disease where the body has produced antibodies against its own cholinergic receptors

Analogues cont'd…

- Administering an agonist increases the chance of activating what few receptors remain



* Asymetric center

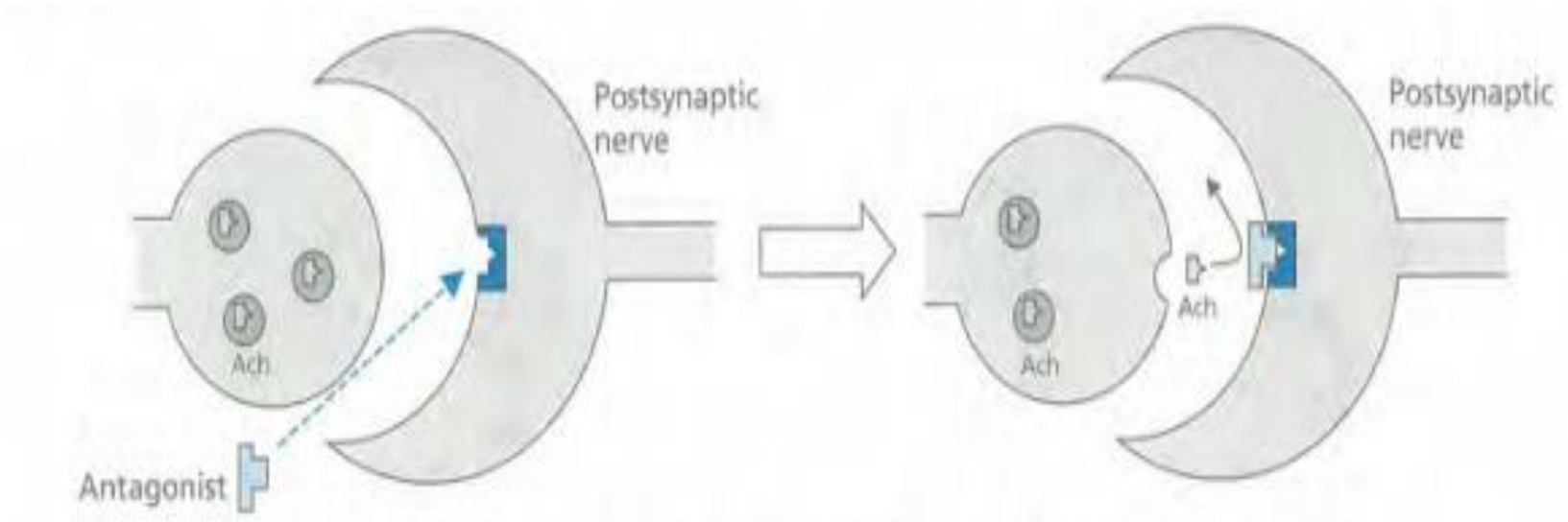
Example of a selective nicotinic agonist

- Selective for nicotinic receptor
- Not used clinically and anticholinesterases are preferred treatments

Antagonists of the muscarinic cholinergic receptor

Actions and uses of muscarinic antagonists

- Bind to receptors but do not activate them
- Inhibit the binding of an agonist



Antagonist cont'd…

- Clinical effects of muscarinic antagonists
  - Reduction of saliva and gastric secretions
  - Reductions of the motility of the GIT and Urinary tract by relaxing smooth muscle
  - Dilatation of eye pupils
  - CNS effects

Clinical uses of muscarinic antagonist

- Ophthalmic examinations
- Shutting down GIT and UT during surgery
- Relief of peptic ulcers
- Treatment of Parkinson's disease
- Treatment of anticholinesterase poisoning
- Treatment of motion sickness

Muscarinic antagonists

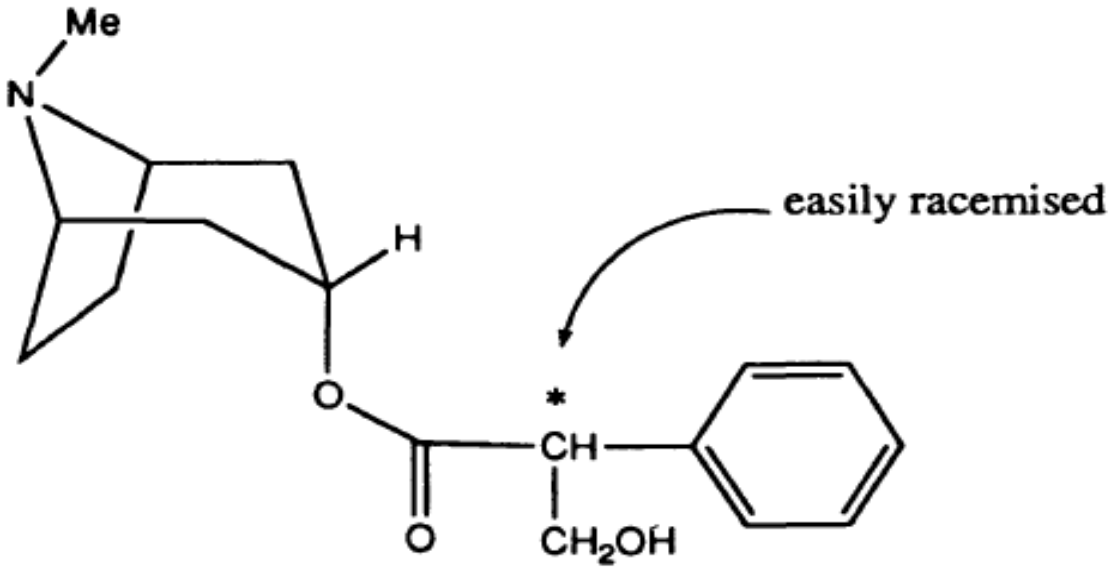
- First antagonists were natural products- in particular alkaloid, nitrogen containing compounds derived from plant

Atropine

- Atropine is present in the roots of *atropa belladonna* (deadly nightshade)
  - ✓Is included in a root extract
- Used by italian women to dilate their eye pupils so that they would appear more beautiful- hence the name belladona
  - Ayshawaria rai bach
- Clinical uses
  - To decrease gastrointestinal motility and
  - to counteract anticholinesterase poisoning

Atropine cont'd…

- Atropine has a chiral center
  - Exists in as racemate
  - It exist as a single enantiomer is called hyoscyamine in plant source

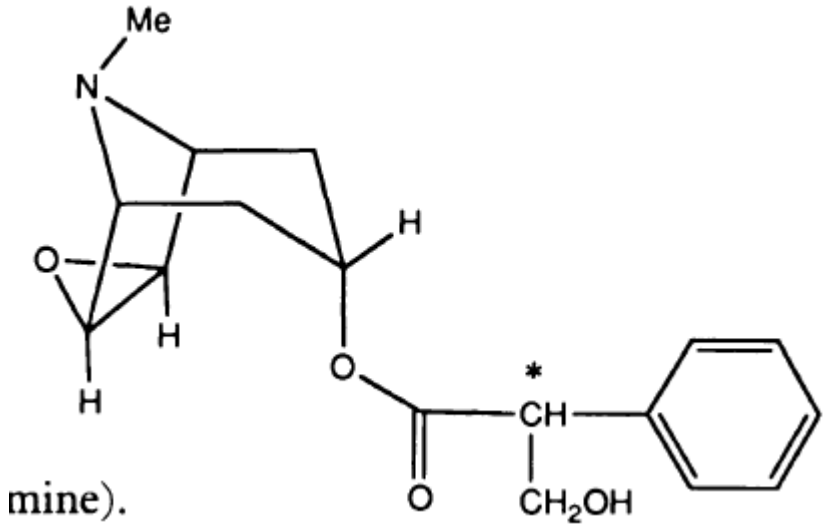


Atropine cont'd…

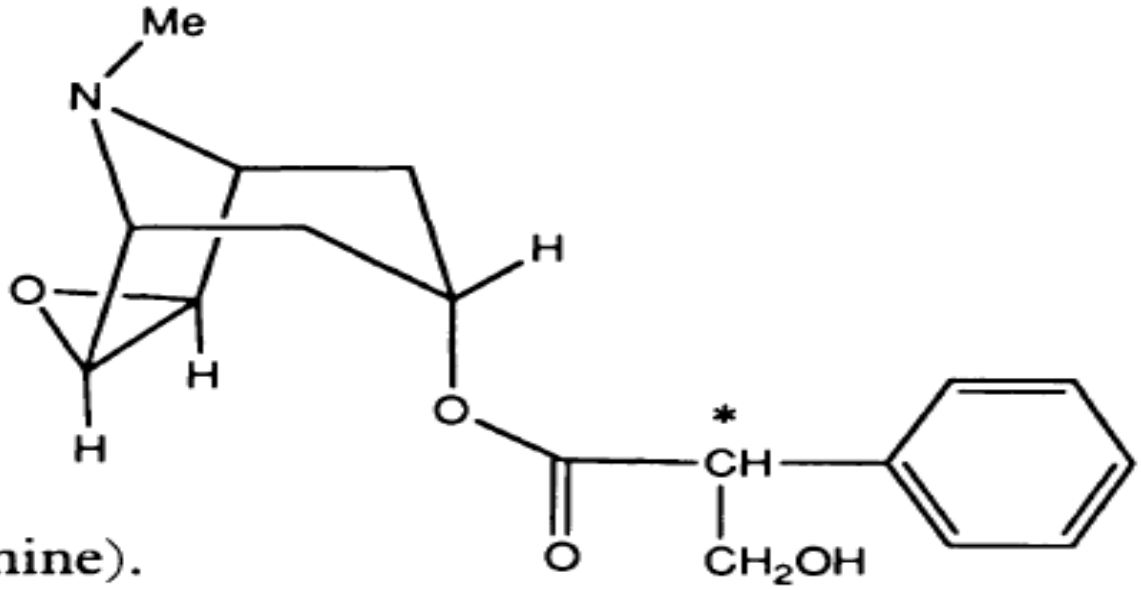
- The asymmetric center is easily racemized because it is next to a carbonyl group
  - This makes the proton attached to the asymmetric center acidic and easily removed

Hyoscine

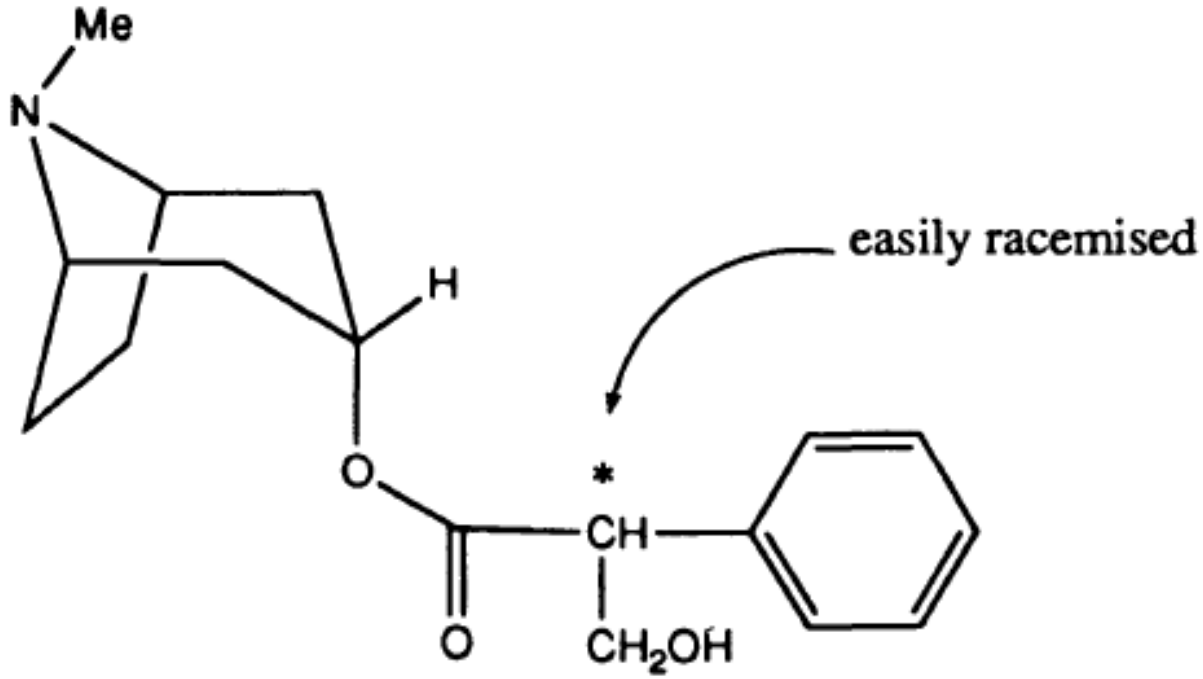
- Hyoscine (scopalamine) is obtained from the thorn apple (*Datura stramonium*)
  - Similar in structure to atropine
  - Used to treat motion sickness



mine).

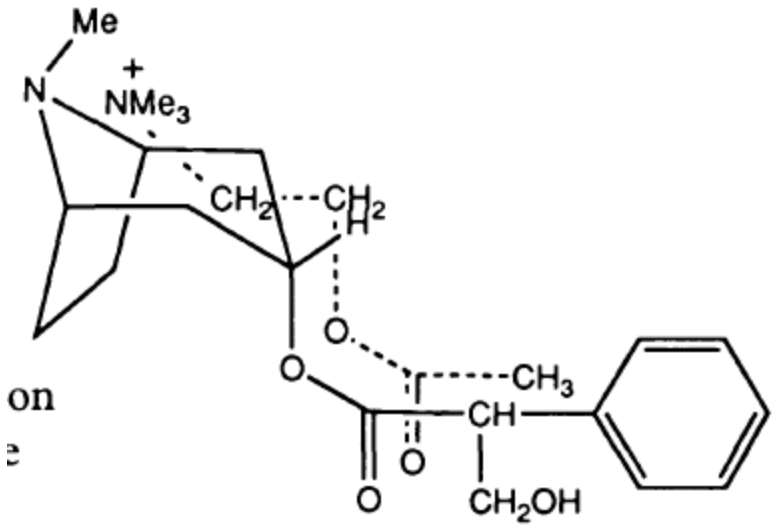


mine).



Atropine cont'd…

- Bind to cholinergic receptor but do not look anything like acetylcholine
  - Contains a basic nitrogen and an ester group
  - Superimposition with acetylcholine shows the distance between ester and nitrogen groups are similar in both structure



Atropine cont'd…

- Nitrogen in atropine is uncharged.
- Implication
  - The nitrogen in atropine must be protonated and charged when it binds to the cholinergic receptor
- Able to bind to the receptor but unable to switch it on? Why?
  - Atropine is a larger molecule than acetylcholine, it is capable of binding to other binding regions outside the acetylcholine binding site

Atropine cont'd…

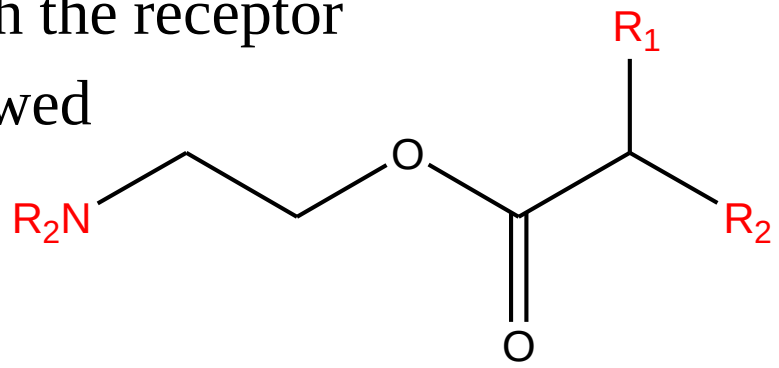
- Both atropine and hyoscine can cross the BBB
  - Because they don't have the quaternary charge, they are only tertiary amines
- Once in the brain they are protonated and antagonize muscarinic receptors in the CNS
- Hallocinogenic activity is brought
  - Both were used by witches centuries ago

Atropine cont'd…

- Truth drug
  - For the interrogation of spies
  - As torture agents with succinylcholine
  - Succinylcholine creates convulsive muscle spasms, inability to breathe, agonizing pain and a living impression of death.
  - Then scopolamine is used to erase this horror so it can be bad when the process was repeated.

SAR for atropine and analogues

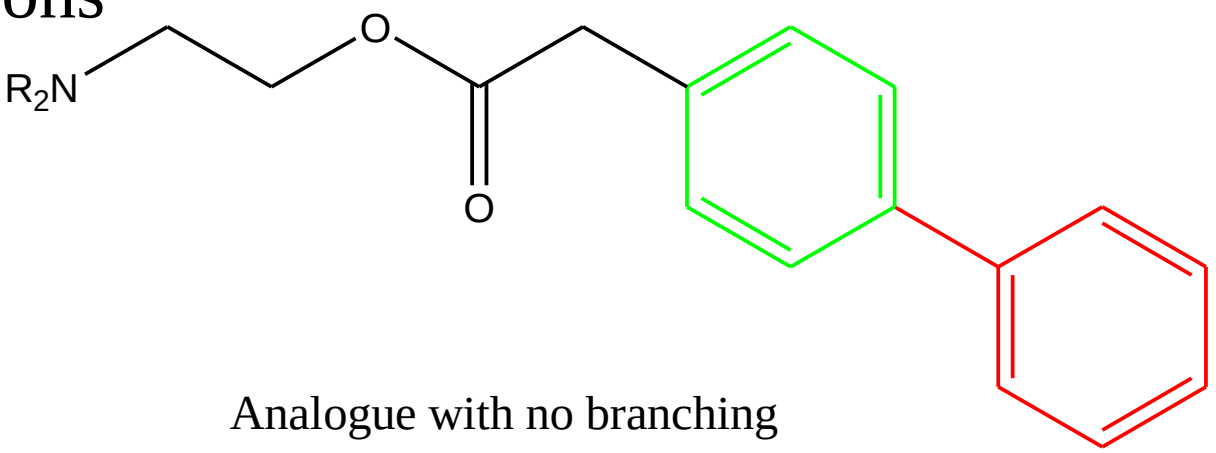
- The alkyl groups (R) on nitrogen can be larger than methyl (in contrast to agonists)
- The nitrogen can be tertiary or quaternary, where agonists must have a quaternary nitrogen.
  - Note, however, that the tertiary nitrogen is probably charged when it interacts with the receptor
- Very large acyl groups are allowed
  - In contrast to agonists



R^1 and R^2= Aromatic or heteroaromatic

SAR cont'd…

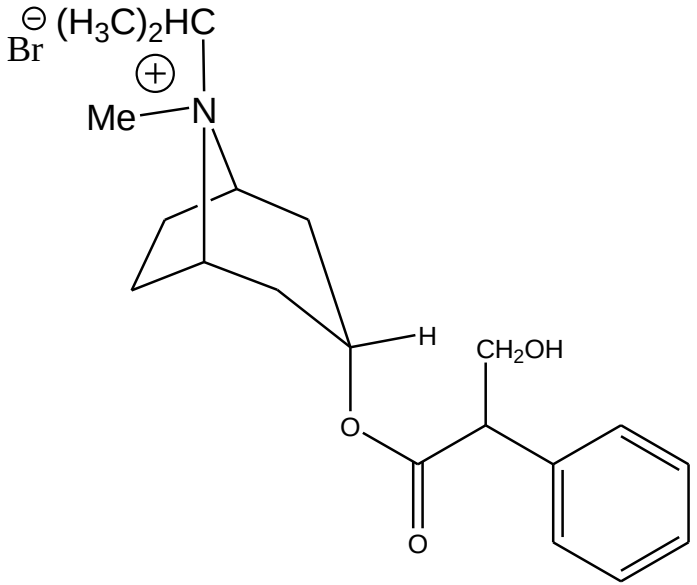
- This big and large acyl groups are necessary for antagonist activity (R1 and R2 can be aromatic or heteroaromatic ring).
  - The bulkiness also has to be arranged in a certain manner;
  - Branching is necessary, Hydrophobic binding regions



Analogue with no branching

Structural analogues based on atropine

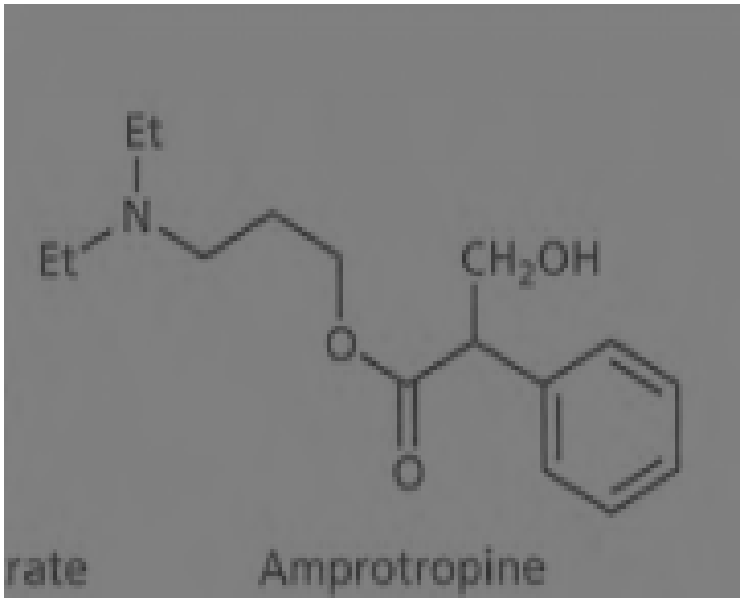
- To reduce CNS effects, quaternary salts of atropine and analogues are used
  - ✓Ipratropium is a bronchodilator
- Useful as an antihistaminic
- Used in the treatment of COPD



Ipratropium

Atropine analogues cont'd…

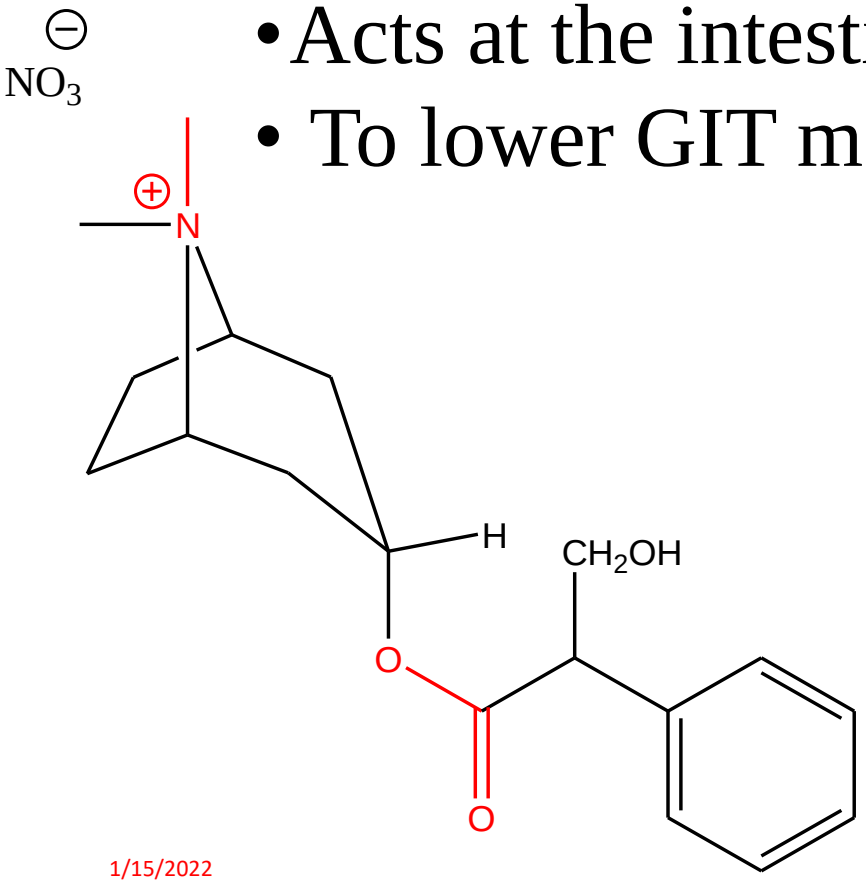
- The complex ring system was not essential for antagonist activity
  - Simplification can be performed
  - E.g. Amprotropine



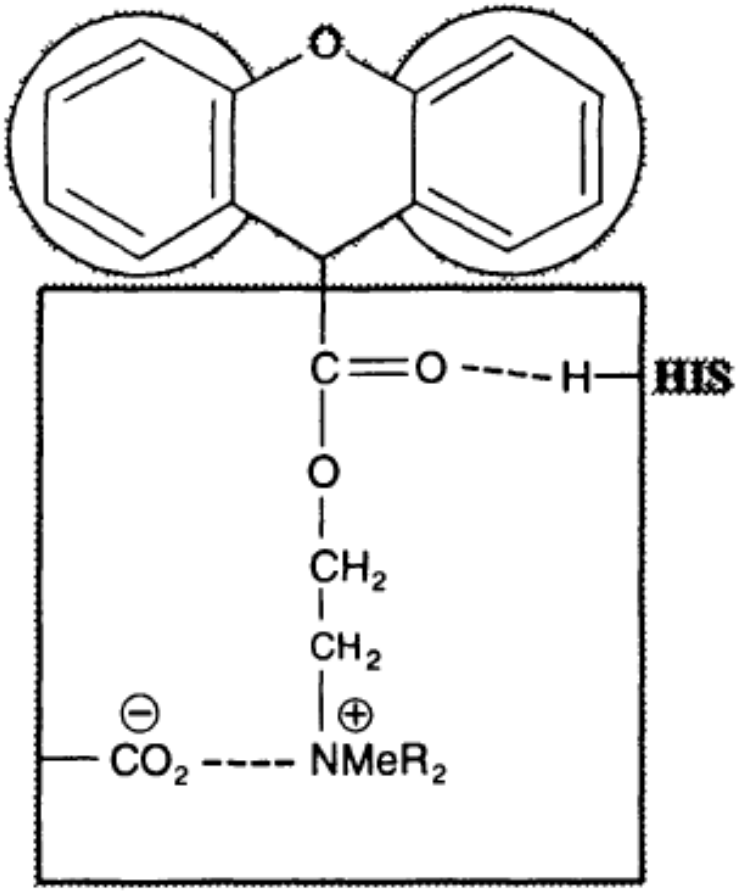
Analogues

Atropine methonitrate

- Acts at the intestine to relieve spasm
- To lower GIT motility



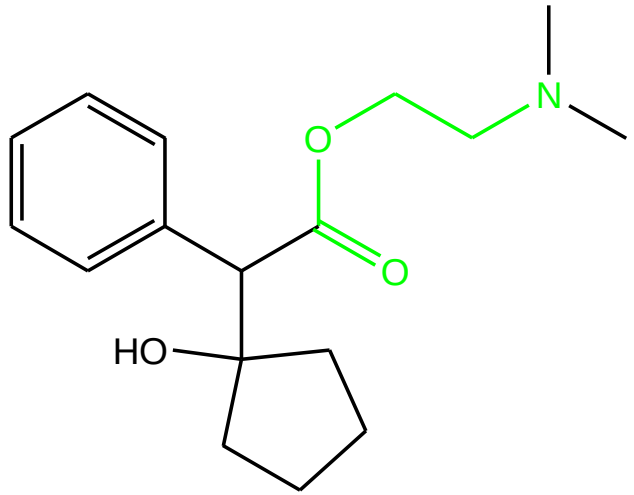
Atropine methonitrate



- Propanthelene which binds to receptor stronger than acetylcholine does

Cyclopentolate

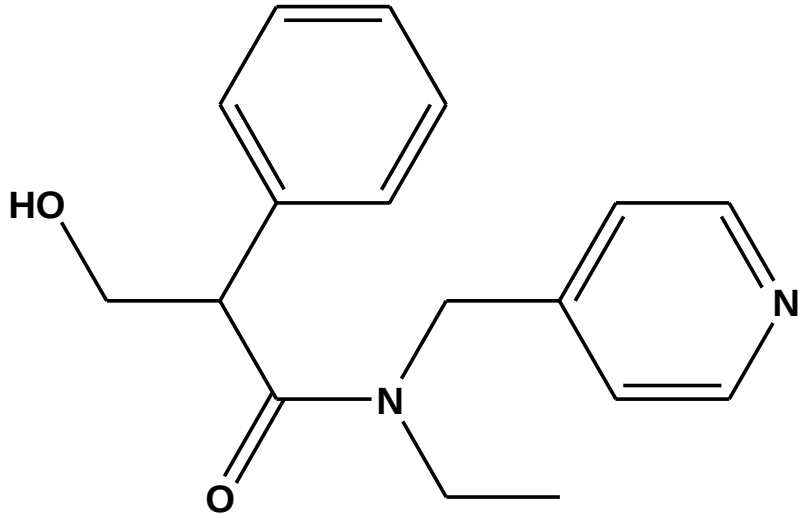
- Used in eye drops to dilate pupils for ophthalmic examination



cyclopentolate

Tropicamide

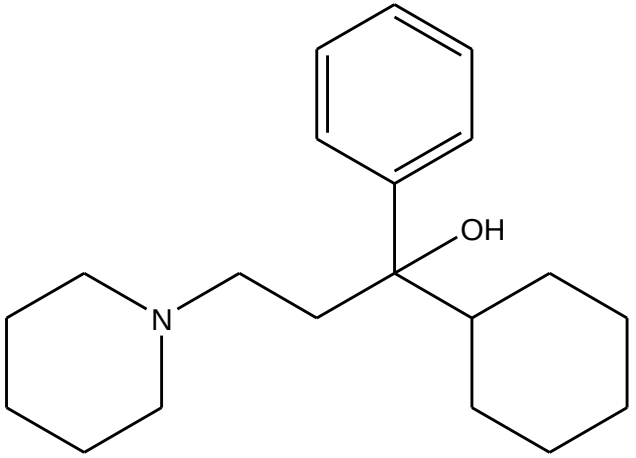
- Same use as cyclopentlate



tropicamide

Trihexyphenidyl

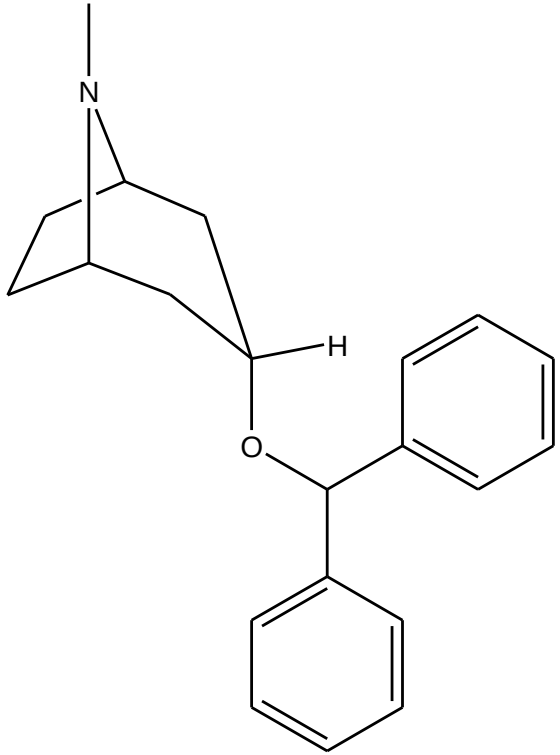
- Anti parkinsonian agent



Trihexyphenidyl

Benzatropine

- Are used centrally to counteract movement disorders due to Parkinson's disease

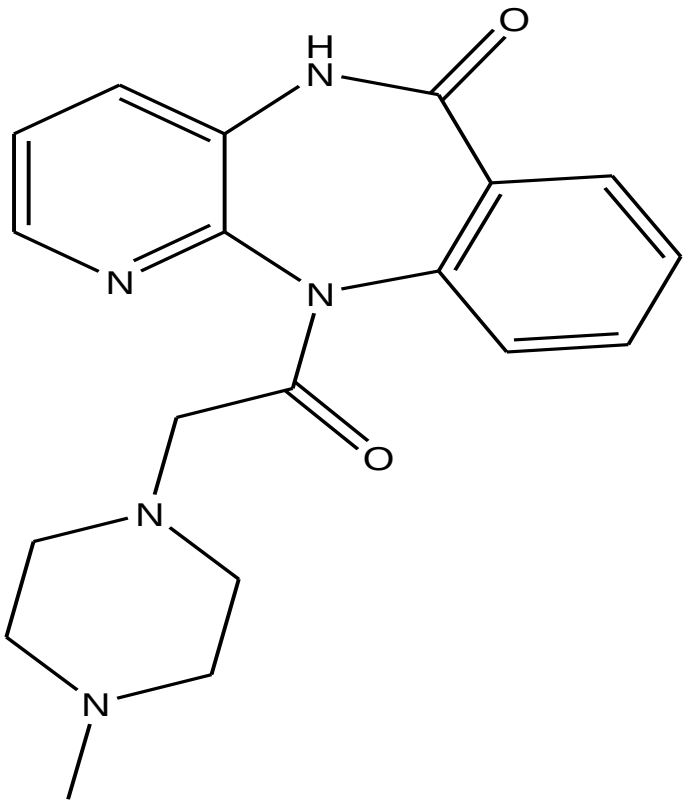


Benzatropine

Analogues cont'd…

Pirenzepine

- Used in the treatment of peptic ulcers
- Is a selective M1 antagonist with no activity against M2 receptors



Pirenzepine

Photoaffinity labeling

- Used for labeling and identification of receptors in tissue preparations

- Antagonist with a radioactive isotope of H or C is used, and the radioactivity reveals where the receptor is located
  - Since antagonists bind more strongly than agonists, they are better compounds to use for labeling and identification of receptors

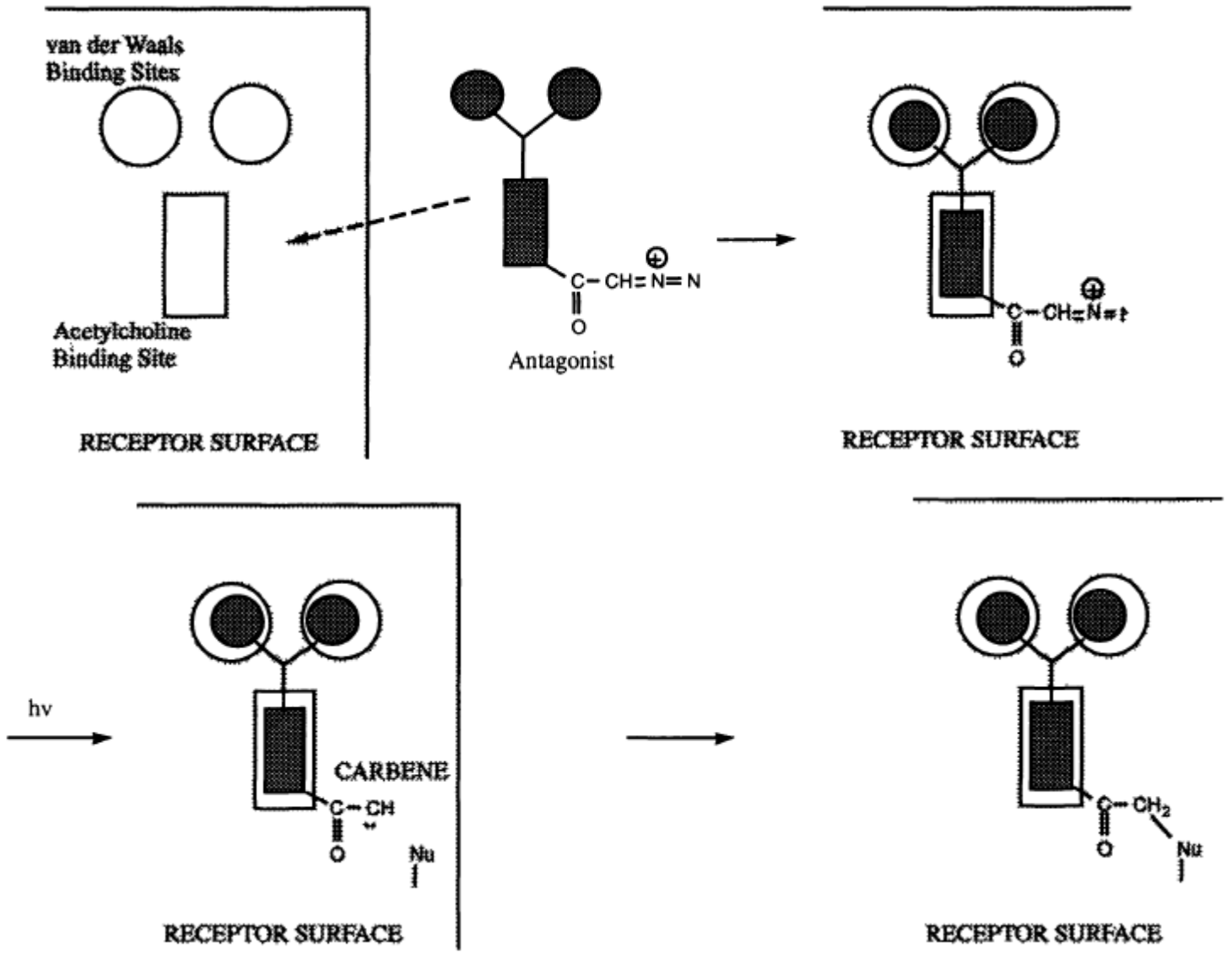
Labeling cont'd…

- Reactive chemical center is incorporated to the chemical center of the molecule
  - Because there are so many sites for the chemical to bind to, the reactive center is only activated by some kind of light or ray

Labeling cont'd…

- Chemical groups such as diazoketones or azides can be converted to highly reactive carbenes and nitrenes respectively, when irradiated

Radio active labeling



- ✓Discuss the role of rigidity
- ✓Explain why acetylcholine is prone to hydrolysis
- ✓Discuss the role of steric shield and electronic effect in producing stable analogues of acetylcholine
- ✓List the clinical use of muscarinic and nicotinic agonists
- ✓Explain the effect of tertiary nitrogen on the activity of pilocarpine
- ✓List the clinical use of both muscarinic and nicotinic antagonists
- ✓Explain the SAR of atropine
- ✓Discuss why atropine can cross BBB and how its inhibited

Antagonists of the nicotinic cholinergic receptor

- Nicotinic receptors are present in nerve synapses at ganglia, as well as at the neuromuscular synapse

  - The drugs are selective because of the distinctive routes taken to reach them not because they have affinity for different receptors

Antagonist cont'd…

- Antagonists of ganglionic nicotinic receptor sites are not therapeutically useful

  - Because they can not distinguish between the ganglia of the sympathetic nervous system and the ganglia of the parasympathetic nervous system; both use nicotinic receptors

Nicotinic antagonists

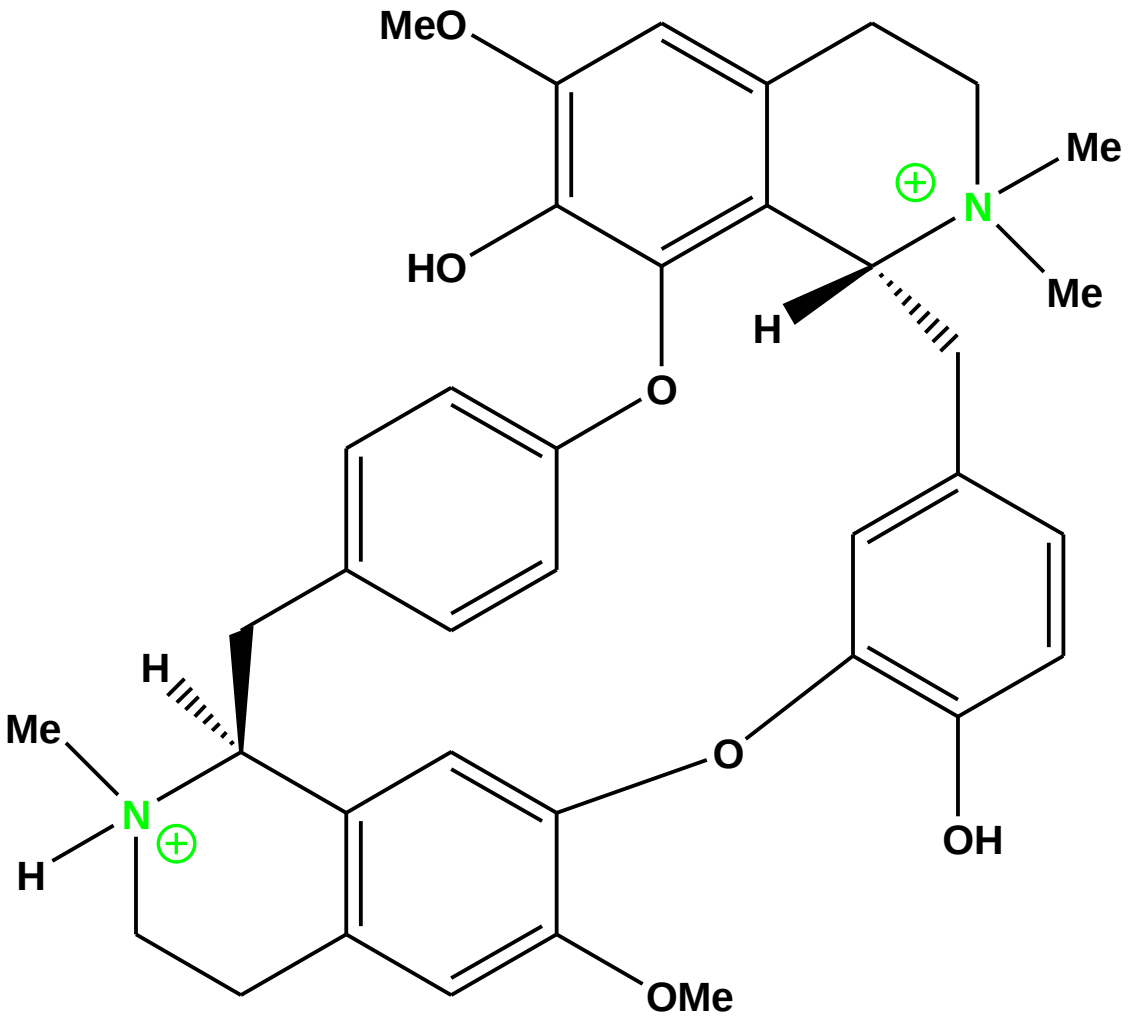
- Curare and tubocurarine
  - Curare was discovered in the sixteenth century
  - Spanish soldiers in south America were under attack by indigenous people using poisoned arrows
  - Used a crude, dried extract from a plant called *chondrodendron tomentosum to* stop the heart and also cause paralysis
    - Antagonist of acetylcholine, blocking nerve transmission from nerve to muscle

Nicotinic cont'd…

- Main therapeutic application is the relaxation of abdominal muscles in preparation for surgery.
  - Allows lower dose of general anesthetic

- Curare is a mixture of compounds
  - In 1935, the active principle tubocurarine was isolated
  - Structure was determined in 1970
  - Used clinically as neuromuscular blocker

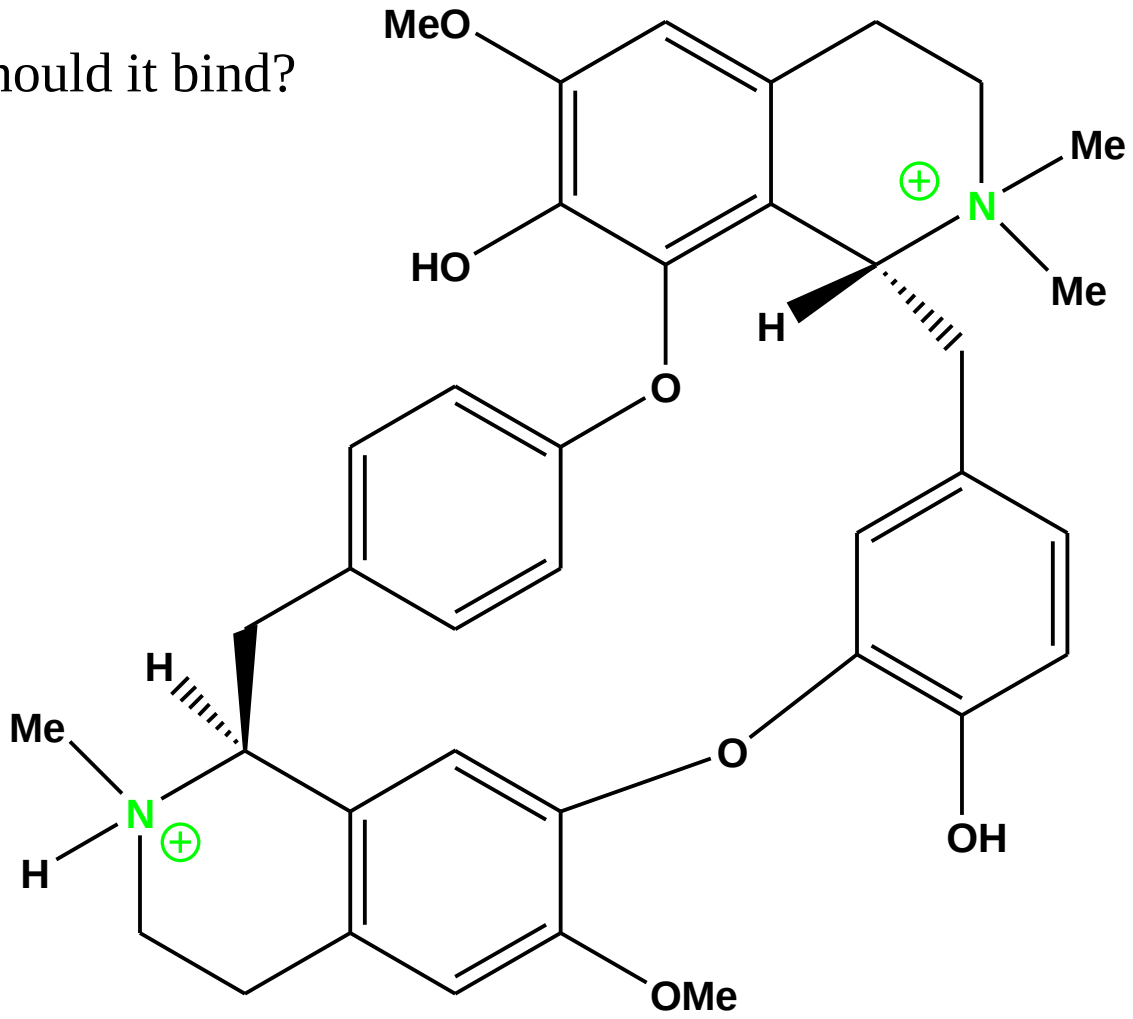
Nicotinic cont'd…



Tubocurarine

- Structure presents problem to theory of receptor binding
  Has two quaternary nitrogens but no ester group.
  studies shows that for good binding , requires more than just a quaternary nitrogen

- How should it bind?



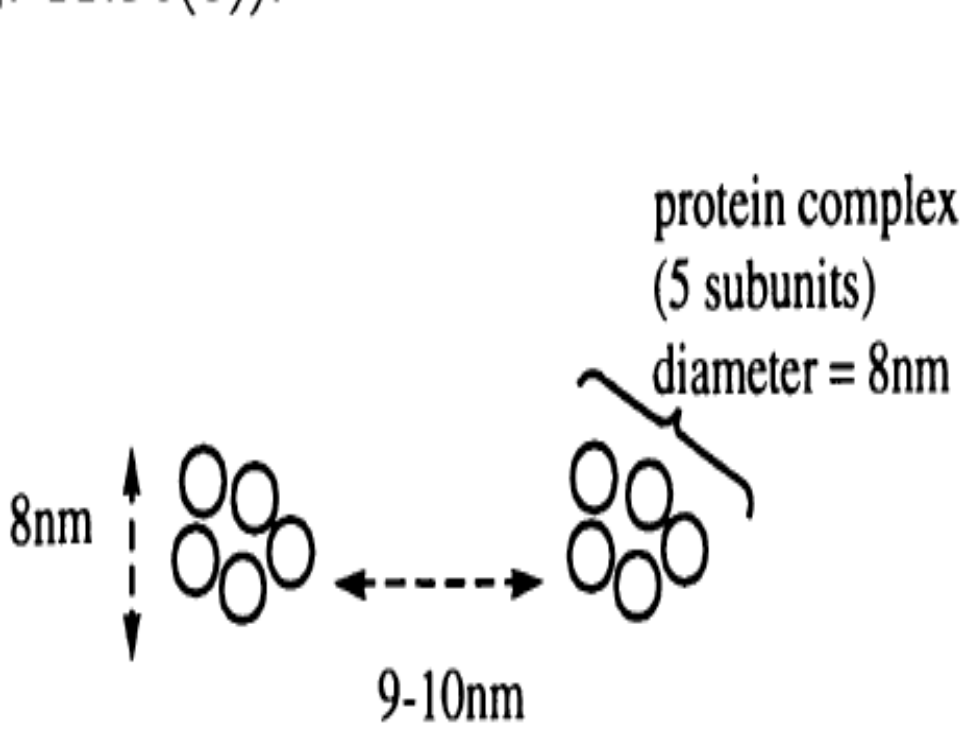
Tubocurarine

Answer

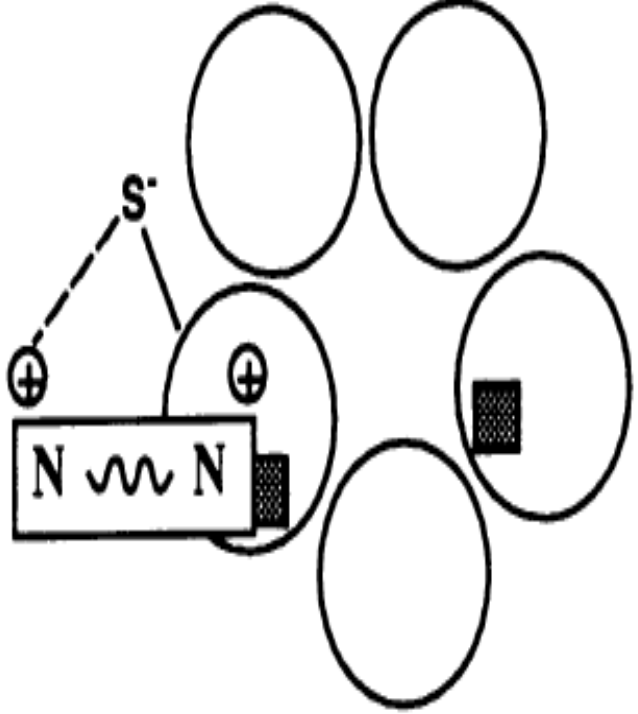
- Molecule has two positively charged nitrogen atoms (one tertiary and one quaternary)

Two possibilities

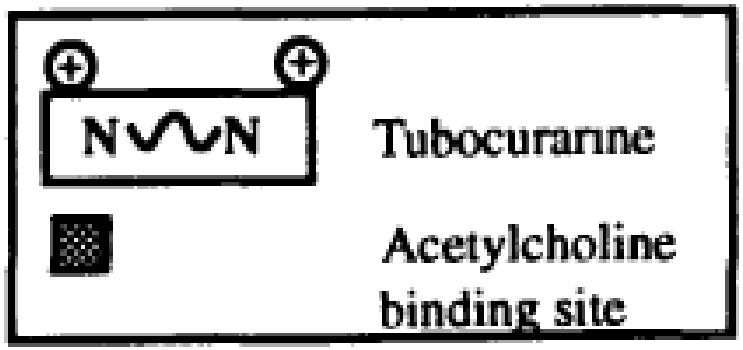
- Originally believed that the distance between the two centers (1.15 nm) might be equivalent to the distance between two separate cholinergic receptors
  - The large tubocurarine molecule could act as a bridge between the two receptor sites
  - Thus spreading a blanket over the two receptors and blocking access to acetylcholine



(a) Receptor Dimer



(b) Interaction with Tubocurarine



Answer cont…

- However the dimensions of the receptor make this unlikely
  - The nicotinic receptor is a protein dimer made up of two identical protein complexes separated by 9- 10 nm- far too large to bridged by the tubocurarine molecule
- The second possibility is that the tubocurarine molecule bridges two acetylcholine binding sites with in the protein complex

Answer cont...

- This appears to be an attractive theory but the two sites are more than 1.15 nm apart and so this too seems unlikely

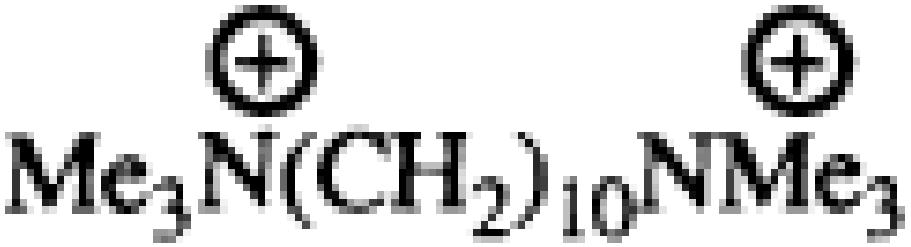
- It has now been proposed that one of the positively charged nitrogens on tubocurarine binds to the anionic binding region of the acetylcholine binding site, while the other binds to a nearby cysteine residue 0.9- 1.2 nm away

Cont…

- It seems highly probable that two ionic binding regions are involved.
  - Such an interaction is extremely strong and would more than make up for the lack of the ester binding interactions
  - It is also clear that the distance between the two positively charged nitrogen atoms is crucial to activity
  - Analogues should fulfill these criteria

Analogues

- Decamethonium and suxamethonium
- Decamethonium is a simple analogue of tubocurarine
  - Is a straight chain molecule and as such is capable of a large number of conformations



Analogues cont'd…

- The fully extended conformation places the nitrogen atoms 1.4 nm apart
- bond rotations can result in other conformations that position the nitrogen centers 1.14 nm apart
  - which compares well with the equivalent distance in tubocurarine (1.15 nm)
- The drug binds strongly to cholinergic receptors and has proved a useful clinical agent

Disadvantages

- Initially it act as an agonist rather than an antagonist
  - Meaning, switches on receptors and leads to muscle contraction
  - After this though, remains bound to the receptor, blocking acetylcholine binding and acting as an antagonist
- It binds too strongly, so patients take a long time to recover from its effects.

Disadvantages cont'd…

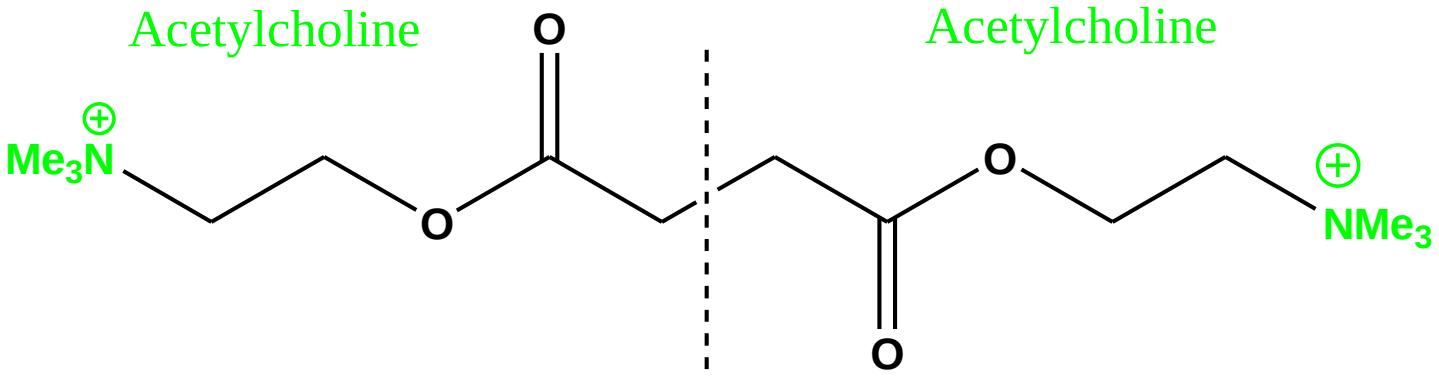
- Both decamethonium and suxamethonium have effects on the autonomic ganglia
  - This explains their side effect
- Decamethonium also lacks total selectivity for the neuromuscular junction and has an effect on choloinergic receptors in the heart.
  - Leading to increased heart rate and a fall in blood pressure

Designing of analogues

- Apart from the stabilization in acetylcholine,
  - A sort of "timer control" where by the molecule can be inactivated more quickly

Suxamethonium

- Incorporation of two ester groups into the chain such that the distance between the charged nitrogen remained the same



Suxamethonium

Suxamethonium cont'd…

- The ester groups are susceptible to chemical and enzymatic hydrolysis
  - The molecule can no longer bridge the binding regions on the receptor and is inactivated
    - ✓ Anchimeric assistance

- The ester groups were also introduced such that suxamethonium mimics two acetylcholine molecules linked end on (affinity increased)

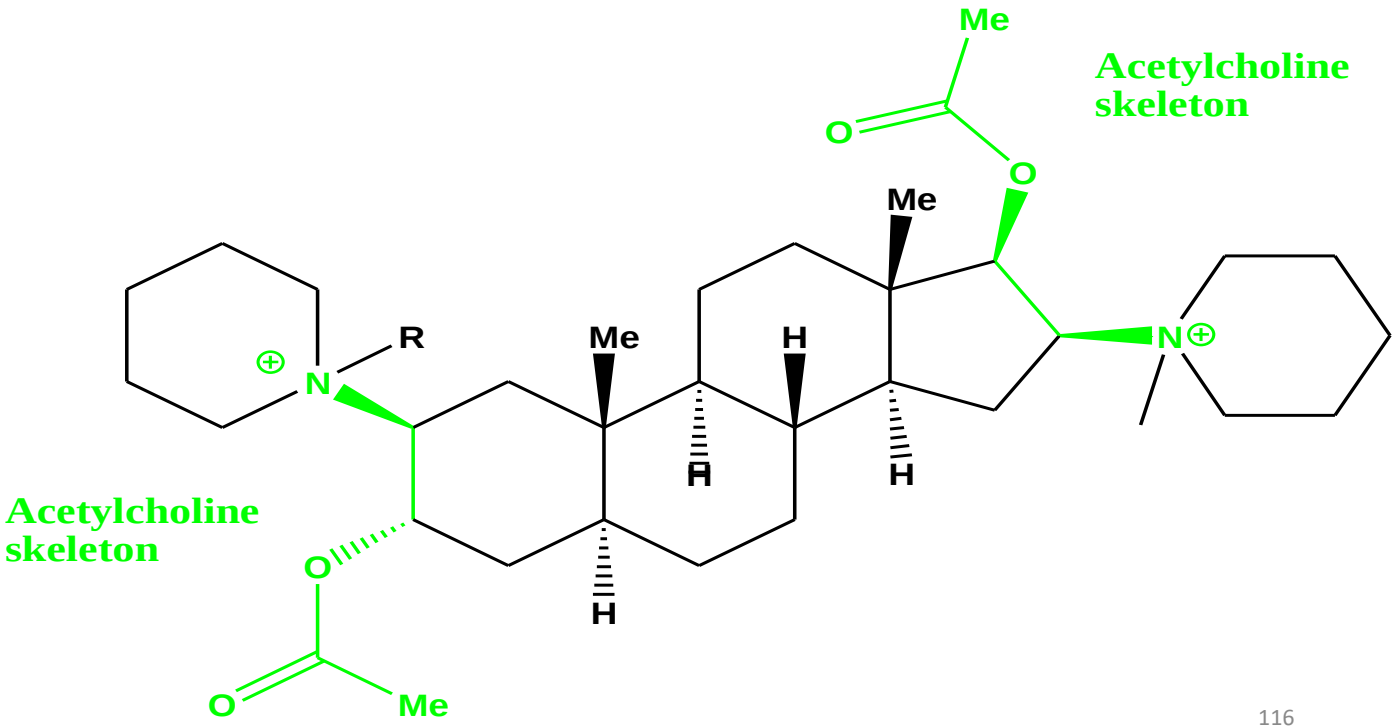
Suxamethonium cont'd…

- Suxamethonium has a fast on set and short duration of actions (5-10 minutes)
- Disadvantages
  - Suffers from various side effects
  - Furthermore, about 1 in 2000 people lacks the enzyme which hydrolyses suxamethonium
- Clinical use
  - In short surgical procedures like insertion of tracheal tubes

Pancuronium and vecuronium

Design was based on tubocurarine, but involved a steroid nucleus acting as a spacer between the two nitrogen groups

- The distance between the quaternary nitrogens is 1.09 nm, compared to 1.15 nm in tubocurarine



Pancuronium (R= Me) and vecuronium (R=H)

Pancuronium and ve cont'd…

- Acyl group was also added to introduce two acetylcholine skeletons into the molecule
  - improve affinity for the receptor sites.
- These compounds have a faster onset of action than tubocurarine
- Do not affect blood pressure

Pancuronium and ve cont'd…

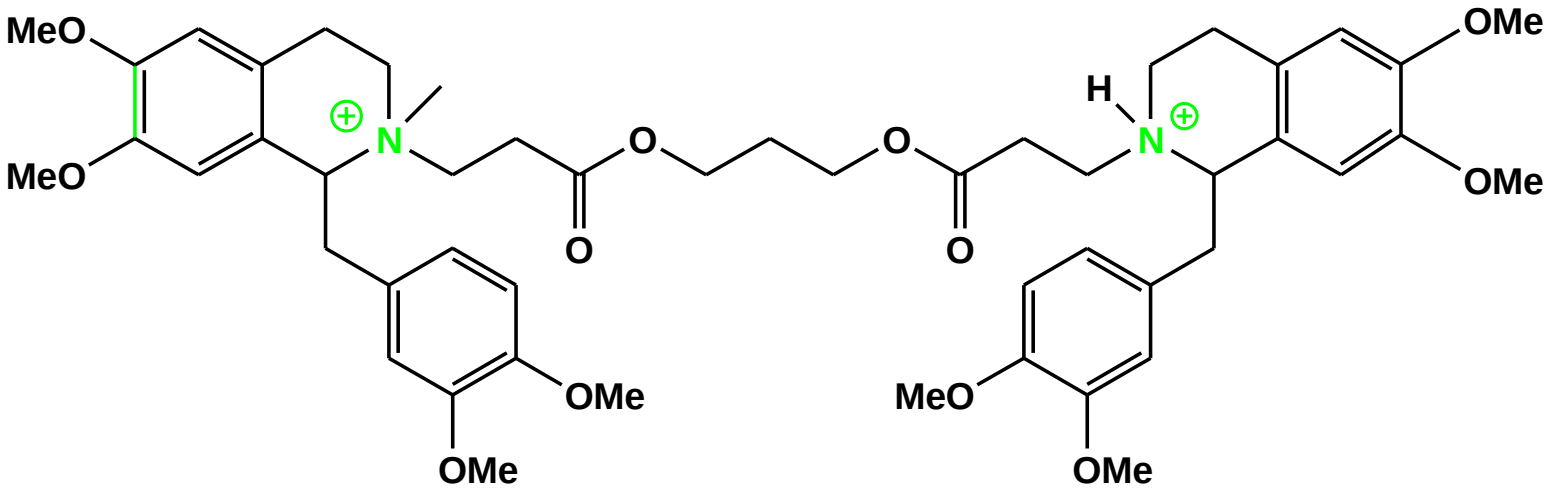
- Advantage
  - They have fewer side effects so used widely clinically
- Disadvantages
  - Not as rapid in onset as suxamethonium
  - Have a longer duration of action

Atracurium and mivacurium

- Atracurium was designed based on the structures of tubocurarine and suxamethonium but
  - It lacks cardiac side effects and is rapidly broken down in blood
  - This rapid breakdown allows the drug to be administered as an intravenous drip

Atracurium and miva cont'd…

- Rapid breakdown was designed into the molecule by incorporating a self-destruct mechanism

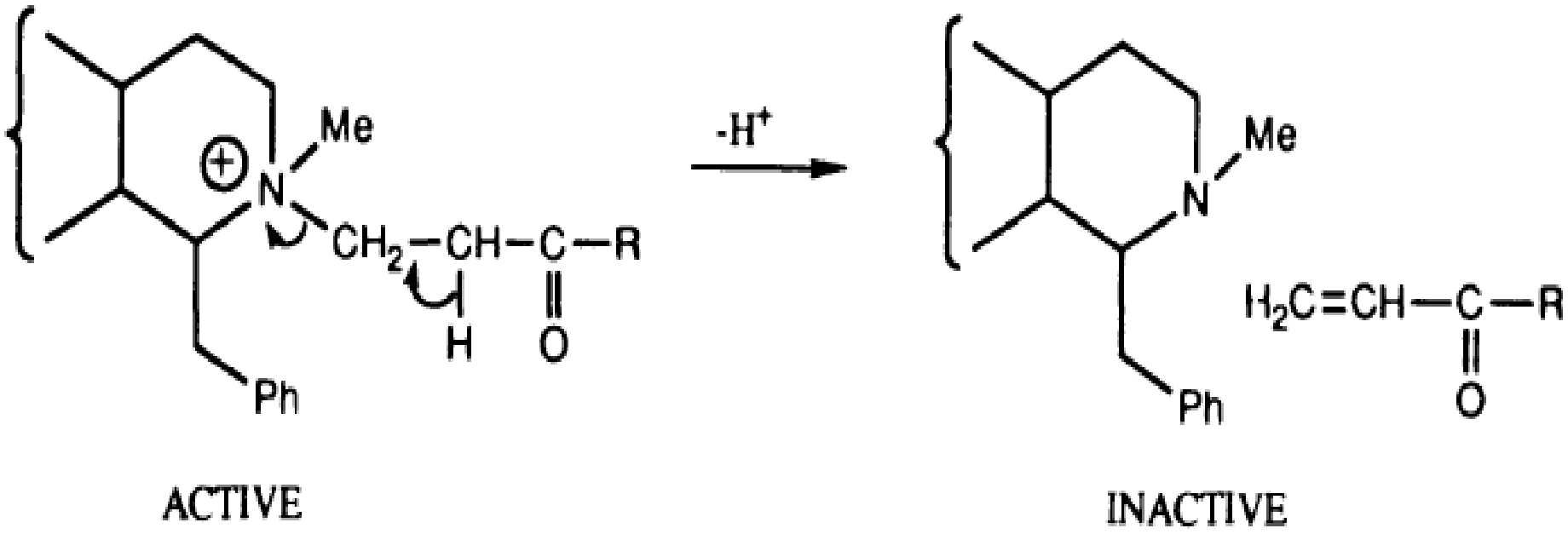


Atracurium

Atracurium and miva cont'd…

- At blood pH, the molecule can undergo a Hoffmann elimination
- Once this happens, the compound is inactivated
  - The positive charge on the nitrogen is lost and the molecule is split in two.
- Clever example of medicinal chemistry drug design in that the very element responsible for the molecule's biological activity promotes its deactivation

Hoffman elimination



ACTIVE

INACTIVE

Important features of Atracurium

- The spacer
  - The 13-atom chain which connects the two quaternary centers
- The blocking units
  - The cyclic structures at either end of the molecule which block the binding site from acetylcholine

Cont..

- The quaternary centers
  - Essential for receptor binding. If one is lost through Hofmann elimination, the binding interaction is too weak and the antagonist leaves the binding site

Cont…

- The Hofmann elimination
  - The esters groups within the spacer chain are crucial to the rapid deactivation process.
  - Hofmann eliminations normally require strong alkaline conditions and high temperature
    - ✓hardly normal physiological conditions.
  - However, if a good electron withdrawing group is present on the carbon, β to the quaternary nitrogen center, it allows the reaction to proceed under much milder conditions

Cont…

- The electron withdrawing group increases the acidity of the hydrogen on the β-carbon such that it is easily lost.
- The Hofmann elimination does not occur at acidic pH, and so the drug is stable in solution at a pH of 3-4
- Can be stored safely in a refrigerator

Advantages of Atracurium

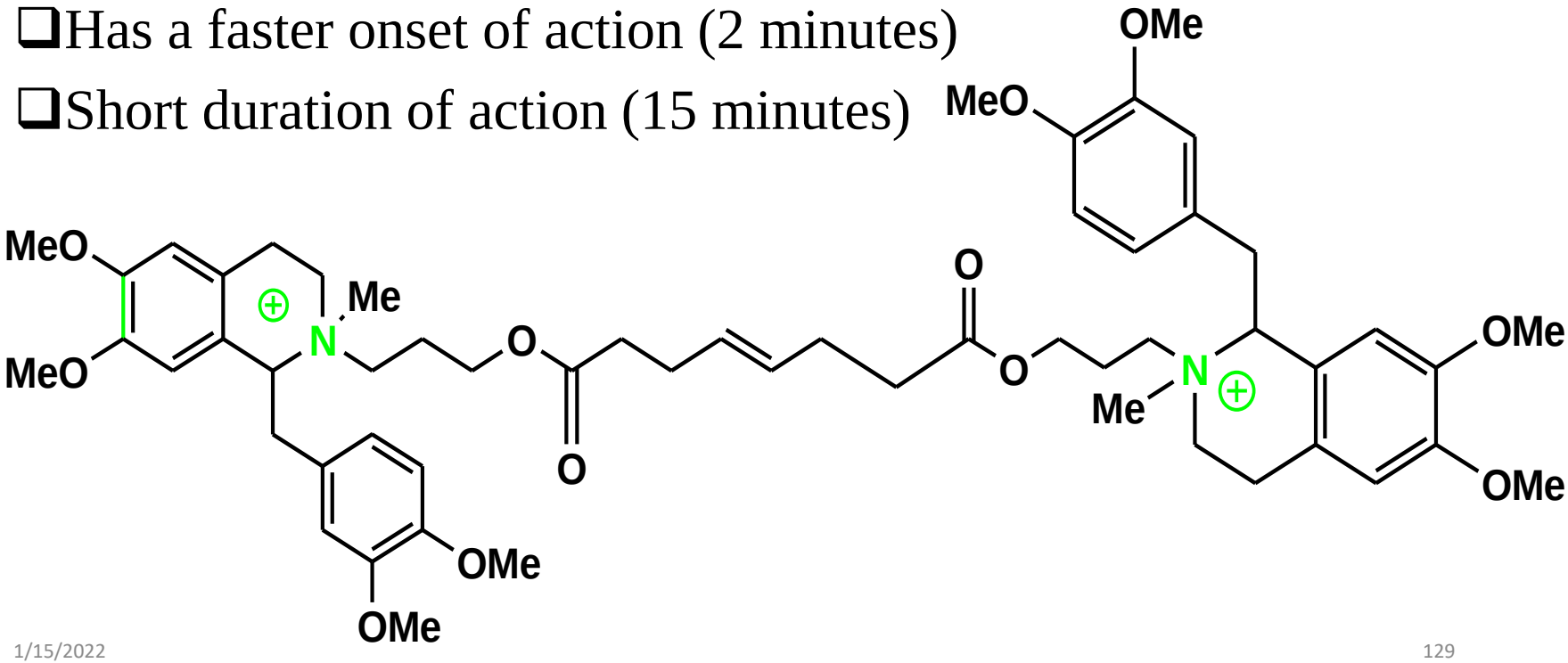
- It can be given through the IV route as long as needed because the drug acts only briefly (30 minutes)
  - As soon as surgery is over, the IV drip can be stopped and antagonism ceases almost instantaneously
- The drug does not require enzymes to become deactivated
  - Deactivation occurs at a constant rate between patients

Cont…

- With earlier agents, deactivation depended on metabolic mechanisms involving enzymic deactivation and/or excretion
  - Efficiency of these processes varies from patient to patient and is particularly poor for patients with kidney failure or with low levels of plasma esterases

Mevacurium

- Is a newer drug similar to atracurium
- Is rapidly inactivated by plasma enzymes as well as by the Hoffman elimination
- Has a faster onset of action (2 minutes)
- Short duration of action (15 minutes)

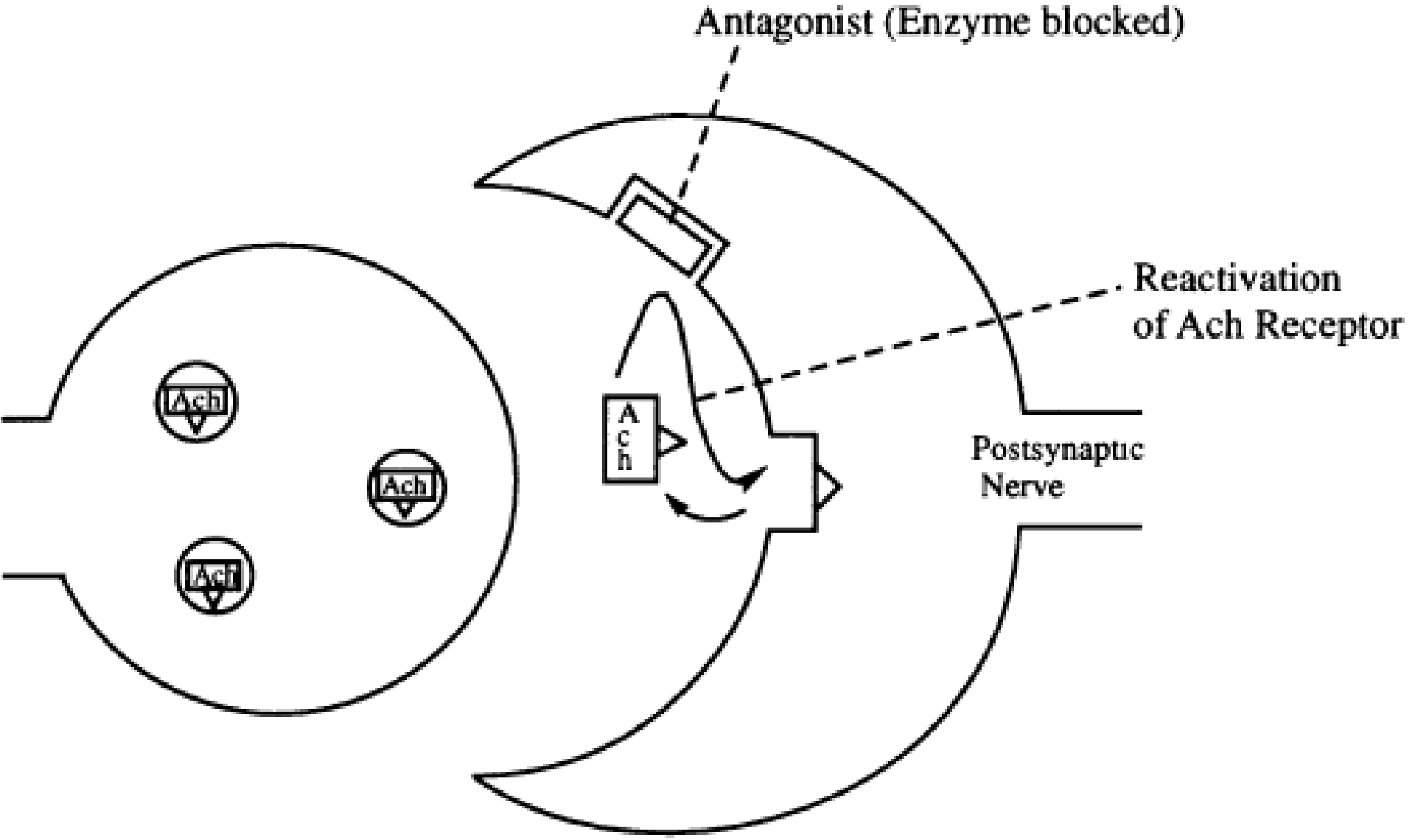


The end!!

Anticholinesterases and acetylcholinesterase

- Effect of anticholinesterases
  - Are antagonists of the enzyme acetylcholinesterase-the enzyme which hydrolyzes acetylcholine
  - With anticholinesterases, the effect is to increase level of acetylcholine and to increase cholinergic effects

Effect of anticholinesterases

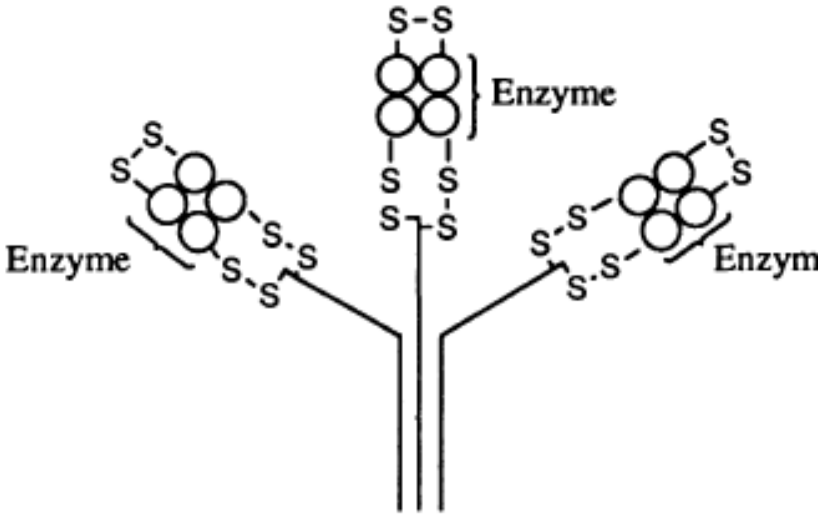


Structure of the acetylcholinesterase enzyme

- Fascinating tree like structure
- Trunk of the tree is a collagen molecule anchored to the cell membrane
- There are three branches leading off from the trunk, each of which hold the acetylcholinesterase

enzyme above the surface

of membrane



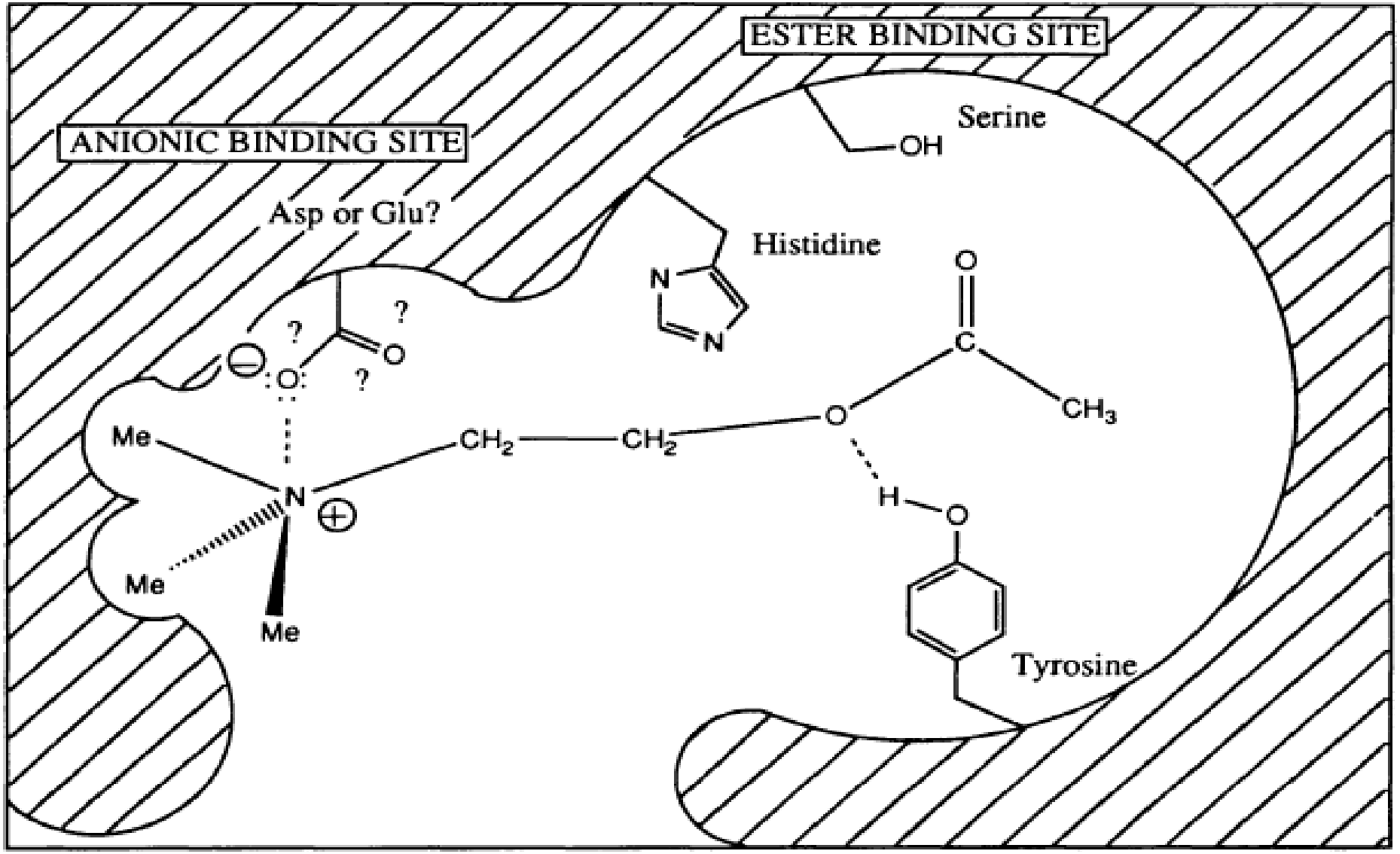
Cont…

- The enzyme is made up of four protein subunits, each of which has an active site
  - One enzyme tree has twelve active sites
- The trees are rooted immediately next to the acetylcholine receptors so that they will efficiently capture acetylcholine molecules as they depart their receptor
  - Acetylcholinesterase is one of the most efficient enzymes

The active site of acetylcholinesterase

- Binding interactions at the active site
  - Acetylcholine binds to the cholinesterase enzyme by
    - ✓Ionic bonding to an Asp residue
    - ✓Hydrogen bonding to a tyrosine residue
  - The aspartate, histidine and serine residues at the active site are involved in the mechanism of hydrolysis
  - The anionic binding site in acetylcholinesterase is similar to the cholinergic binding site

Binding interactions at the active site



Mechanism of hydrolysis

- The histidine residue acts as an acid base catalyst throughout the mechanism
  - While serine plays the part of nucleophile
  - Serine (aliphatic alcohol) is only a poor nucleophile but the nearby histidine provides acid/ base catalysis

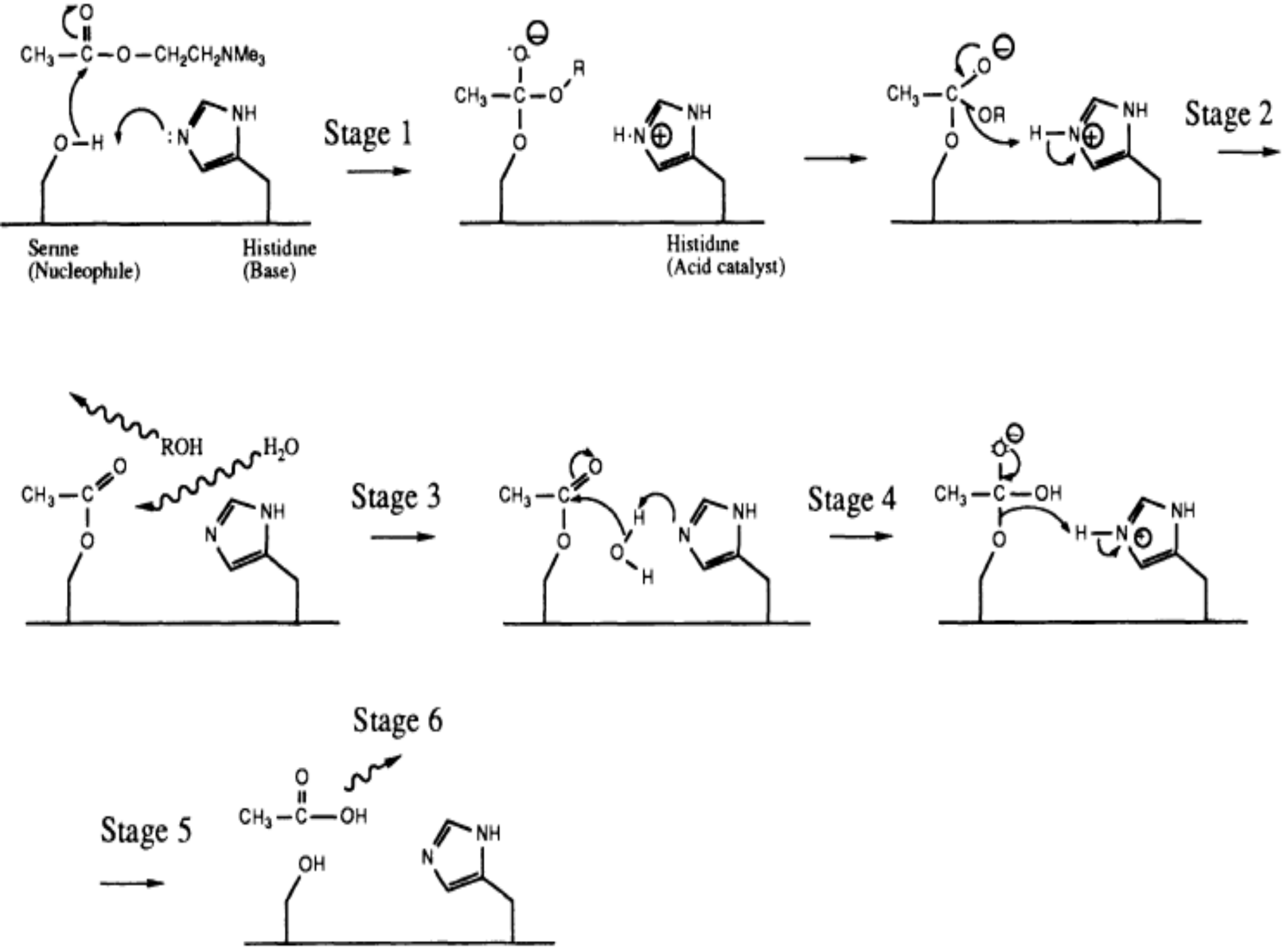
Stages of mechanism

- Stage I
  - Acetylcholine binds to the acetylcholinesterase.
  - The Histidine residue acts as a base to remove a proton from the serine hydroxyl group, thus making it strongly nucleophilic
  - Nucleophilic substitution to the ester group takes place and opens up the carbonyl group

Cont…

- Stage II
  - The carbonyl group reforms and expels the alcohol portion of the ester (i.e. choline)
  - This process is aided by histidine which now acts as an acid catalyst by donating a proton to the departing alcohol
- Stage III
  - The acyl portion of acetylcholine is now covalently bound to the site
  - Choline leaves the active site and is replaced by water

- Stage IV:
  - Water is normally a poor nucleophile, but once again histidine acts as a basic catalyst and nucleophilic addition takes place, once more opening up the carbonyl group
- Stage V:
  - The carbonyl group is reformed and the serine residue is released with the aid of catalysis from histidine
- Stage VI:
  - Ethanoic acid leaves the active site and the cycle can be repeated


Stage 1
Stage 2
Stage 3
Stage 4
Stage 5
Stage 6
Serine
(Nucleophile)
Histidine
(Base)
Histidine
(Acid catalyst)
ROH
H_2O

Cont…

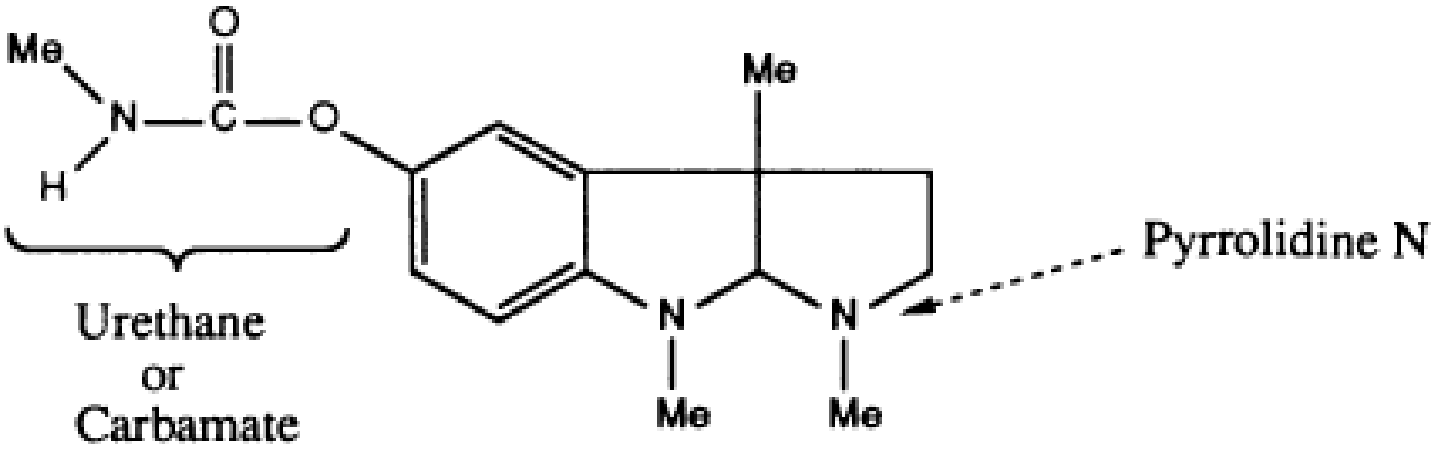
- Efficient enzyme
  - 100,000 X faster than chemical hydrolysis
- Faster than chemical hydrolysis

Anticholinesterase drugs

- Anticholinesterase drugs are antagonists
- This inhibition can be either reversible or irreversible depending on drug interaction with the active site
- Two main groups of anticholinesterases
  - The carbamates
  - The organophosphorus agents

The carbamates

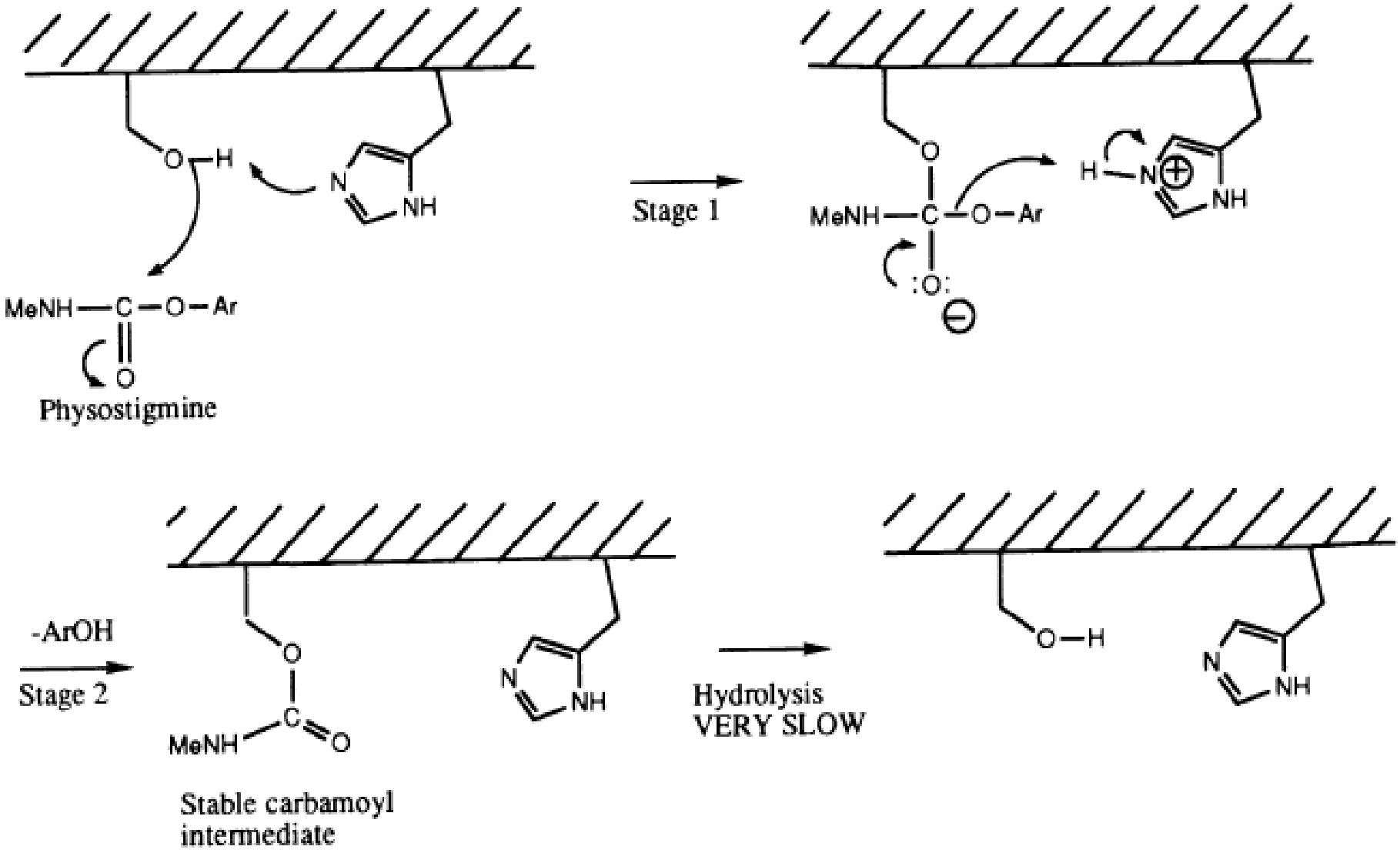
- Natural product provided the lead compound
- The natural compound was Physostigmine discovered in 1864 as a product of the poisonous calabar bean from west Africa
  - Structure established in 1925



Structure-Activity Relationship

- The carbamate group is essential for activity
  - The carbamate is the crucial group responsible for physiostigmine's antagonistic properties
- The benzene ring is important
  - May be involved with extra hydrophobic binding
  - Important in mechanism of inhibition because it provides a good leaving group
- The pyrollidine nitrogen which is ionized at blood pH is important
  - Positively charged nitrogen binds to the anionic binding region of the enzyme

Mechanism of inhibition



Cont…

- The first three steps proceed as normal with histidine catalyzing the nucleophilic attack of the serine residue on physiostigmine (stages 1 and 2)
- The alcohol portion (this time a phenol) is expelled with the aid of acid catalysis from histidine and the phenol leaves the active site to be replaced by a water molecule (stages 3 and 4)

Cont…

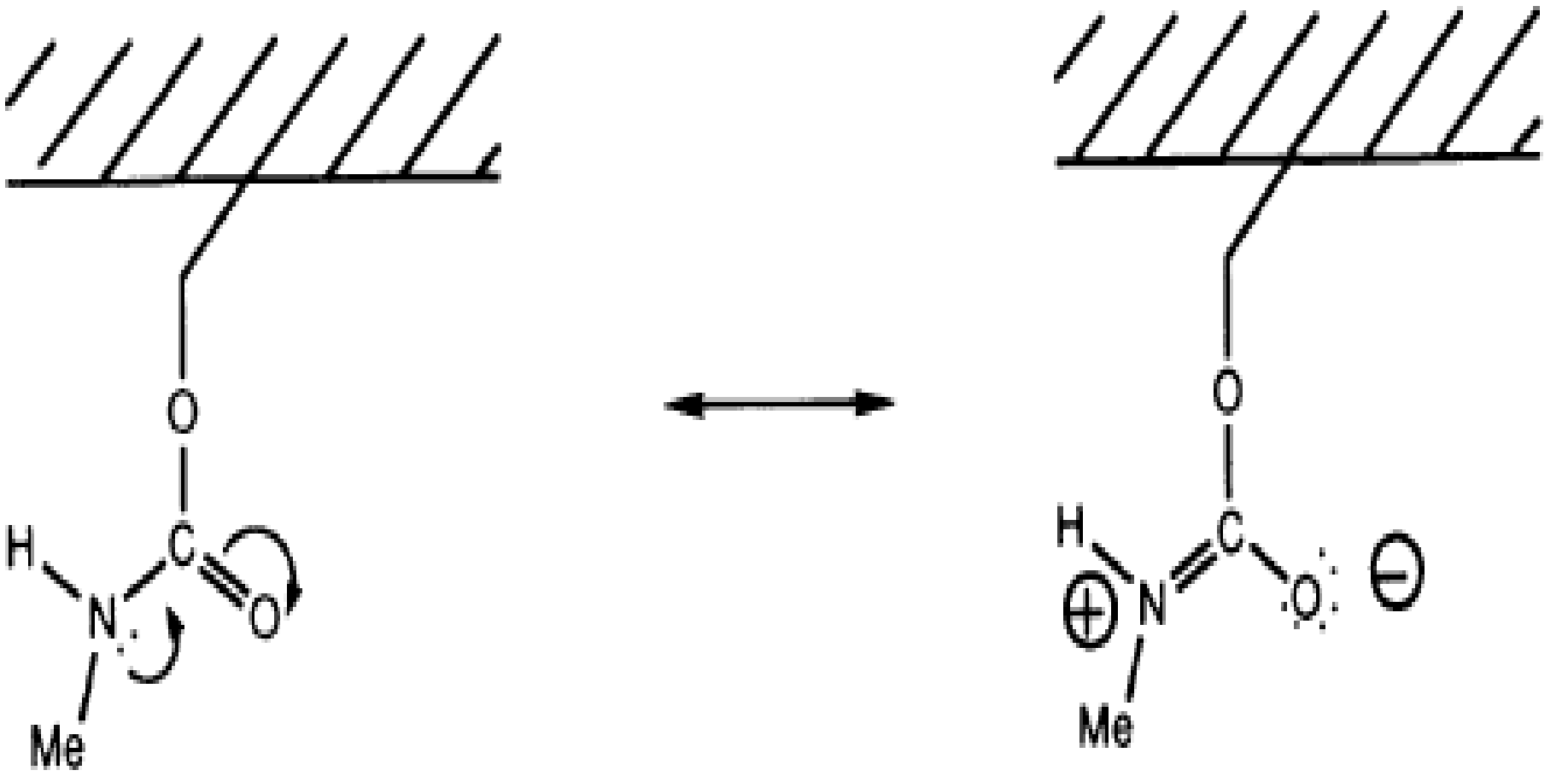
- However, the next step turns out to be extremely slow
  - Is the rate limiting step
  - Water cannot attack the carbamoyl intermediate
- Hydrolysis of physostigmine is forty million times slower than acetylcholine's

- Why is this final stage so slow?

Cont…

- The carbomyl/ enzyme intermediate is stabilized because the nitrogen can feed its lone pair of electrons into the carbonyl group
  - This drastically reduces the electrophilic character and reactivity of the carbonyl group

Stabilization of the cabomyl/ enzyme intermediate

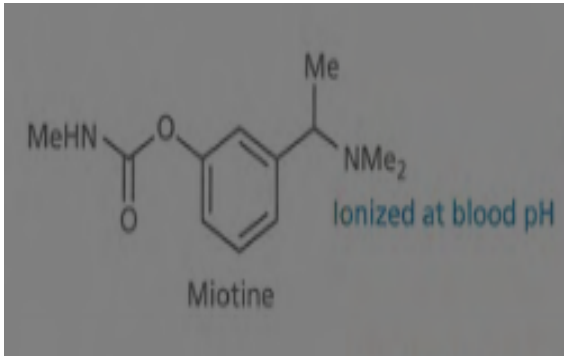


Carbonyl group 'deactivated'

Analogues of physiostigmine

- Physiostigmine has limited medicinal use since it has serious side effects
  - Used only in the treatment of glaucoma and antidote for atropine poisoning
- Simpler analogues, however, have been used in the treatment of myasthenia gravis and as antidote to curare

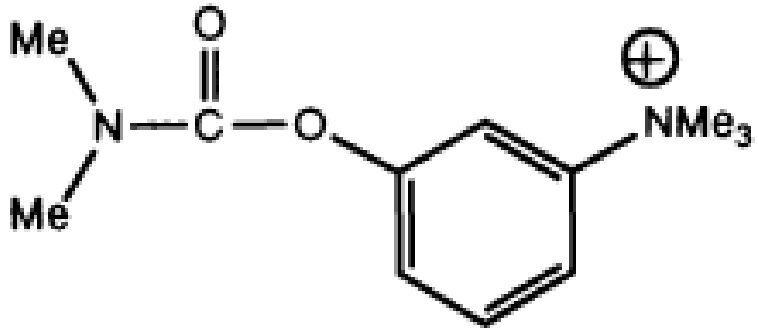
Analogues…



- Miotin has the necessary carbamate, aromatic and tertiary aliphatic nitrogen groups
- Active as antagonist
- Disadvantages
  - It is susceptible to chemical hydrolysis
  - It can cross the blood brain barrier as the free base.
    - This results in side effects due to its action in the CNS

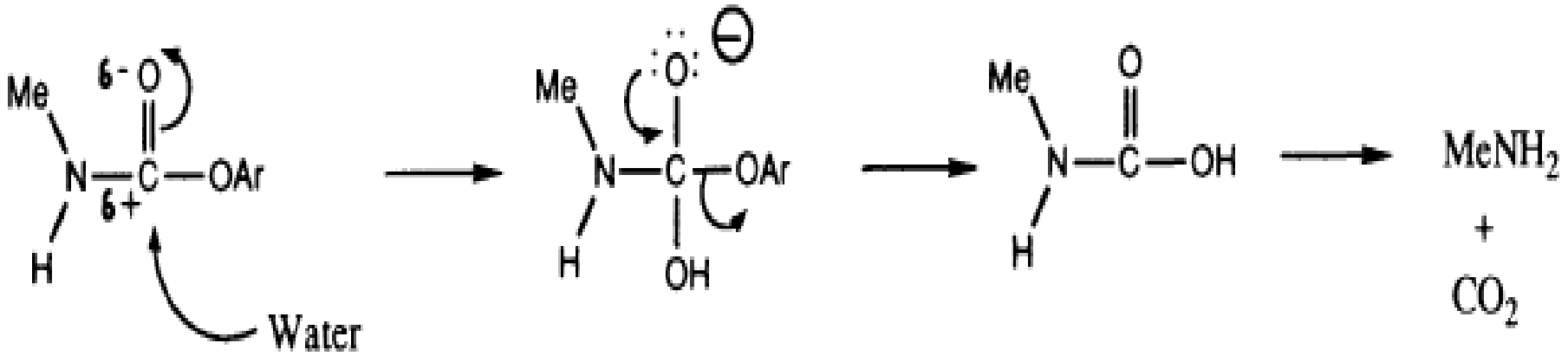
Neostigmine

- was designed to overcome both of the above problems
  - Can not pass the BBB because it is charged analogue
  - Stability to hydrolysis is achieved by using a dimethylcarbamate group rather than a methylcarbamate group



Cont…

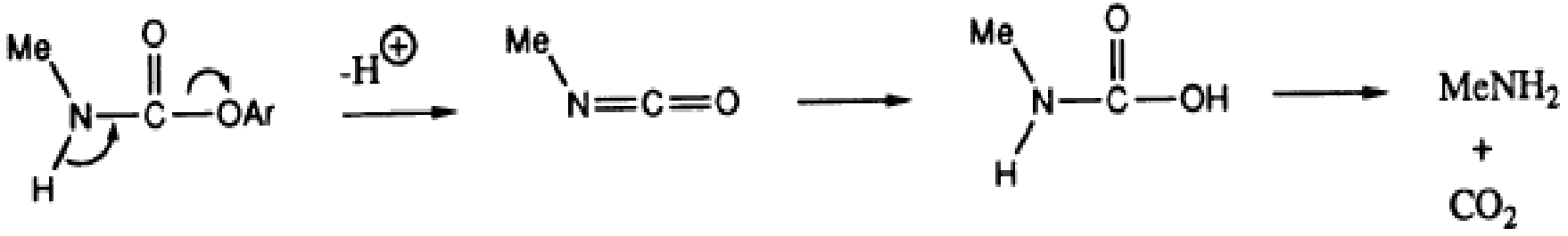
- Two possible mechanisms for hydrolysis



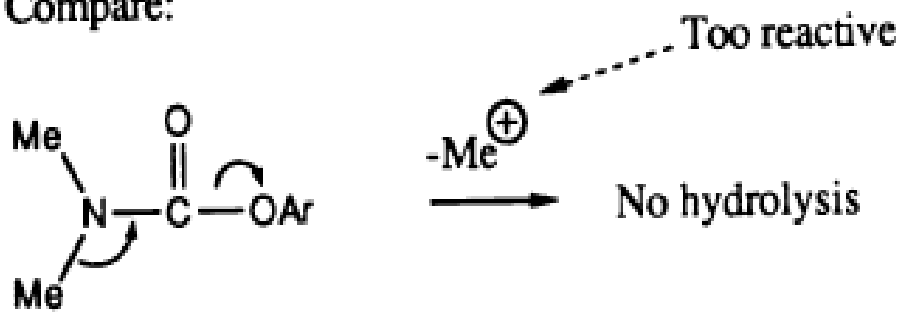
- Rate of reaction depends on the electrophilic character of the carbonyl group and if this is reduced the rate is reduced.
- The lone pair of electrons from the nitrogen tend to stabilize the carbonyl group
  - The presence of a second methyl group on the nitrogen has an inductive “pushing” effect which increases electron density on the nitrogen and further encourages the nitrogen lone pair to interact with the carbonyl group

Mechanism cont…

- Mechanism II is a fragmentation where by the phenolic group is lost before the nucleophile is added
  - The mechanism requires the loss of a proton from a nitrogen.
  - Replacing this nitrogen with a methyl group would severely inhibit the reaction
    - ✓since the mechanism would require the loss of a methyl group- a highly disfavored reaction


Me
O
N
C
OAr
H
-H⊕
Me
N=C=O
Me
O
N
C
OH
H
MeNH2
+
CO2

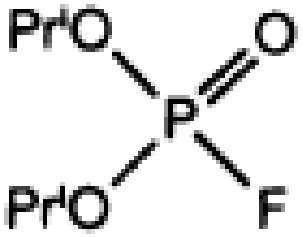
Compare:


Too reactive
Me
O
N
C
OAr
Me
-Me⊕
No hydrolysis

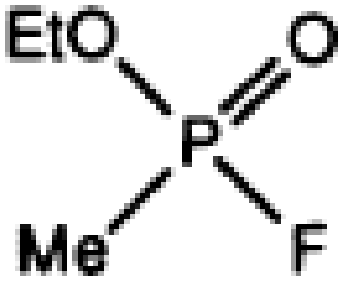
- Two further points about neostigmine
  - The quaternary nitrogen is 4.7 Ao away from the ester group which matches well the equivalent distance in acetylcholine
  - The direct bonding of the quaternary center to the aromatic ring reduces the number of conformations the molecule can make
  - This is an advantage

Organophosphourus compounds

- Used as nerve gas agents in world War II
- In peacetime, they are used as insecticides
- Nerve gases
  - Dyflos and sarin

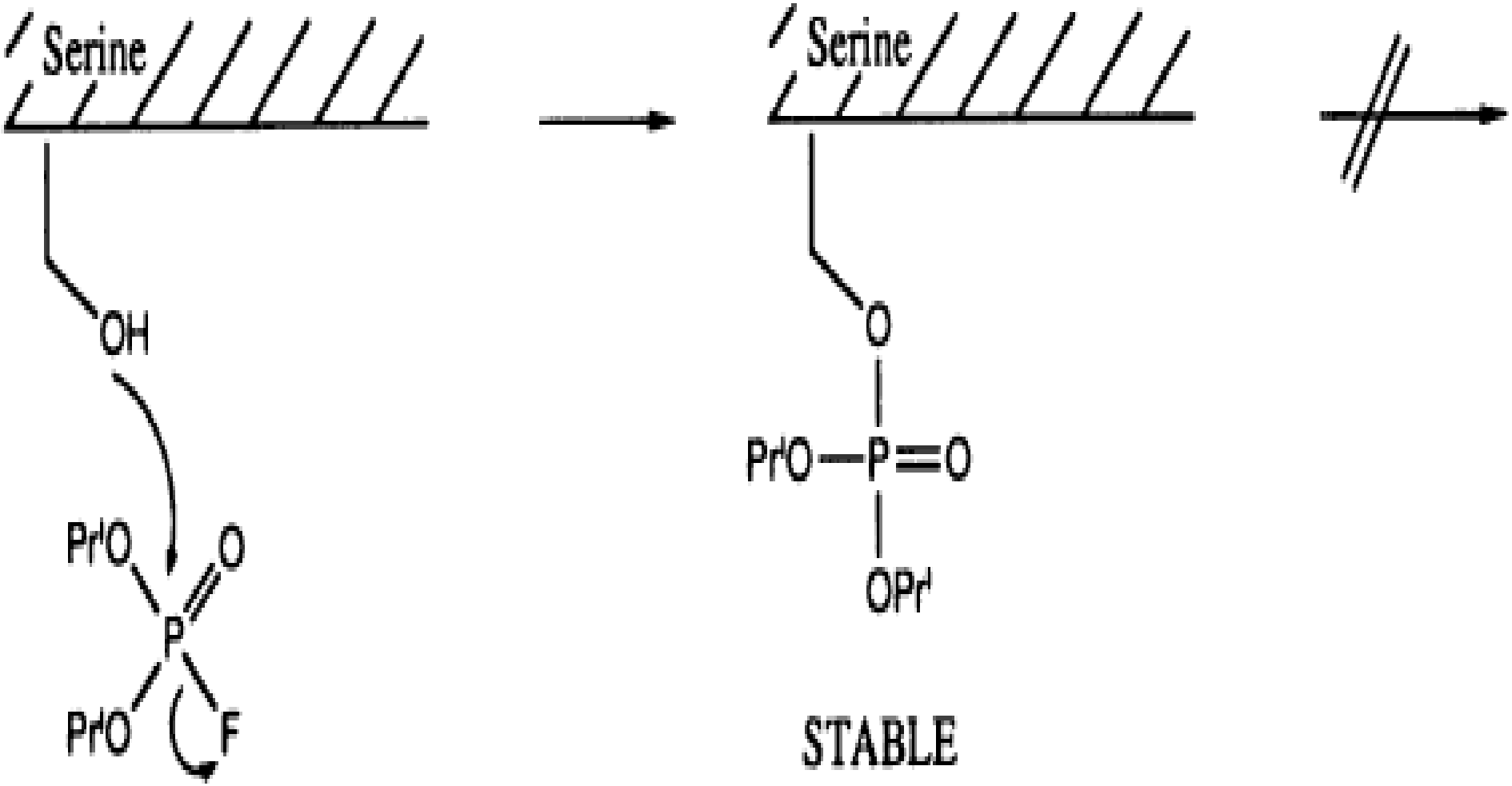


DYFLOS (Diisopropyl fluorophosphonate)



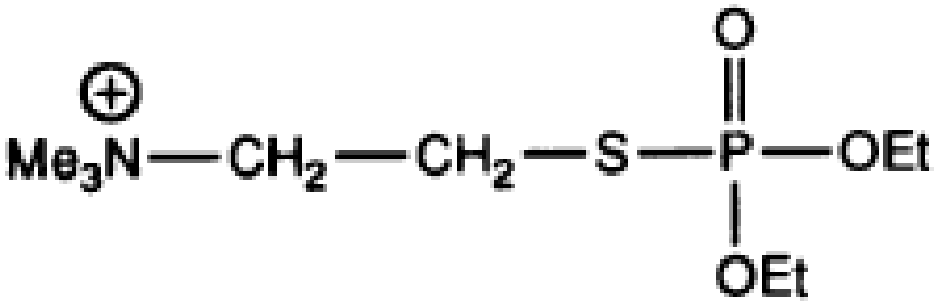
SARIN

Action of dyflos



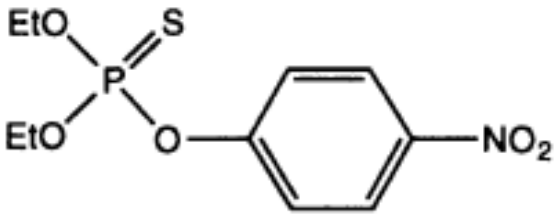
- Dyflos has an LD50 of 0.01 mg/kg
  - Inhibits acetylcholinesterase by irreversibly phosphorylating the serine residue at the active site
  - Its effect is that acetylcholine can not be hydrolyzed hence continuous stimulation of the cholinergic system
  - Permanent contraction of skeletal muscle leading to death

- Ecothiopate was designed to fit the active site more accurately
  - Ecothiopate is used medicinally in the treatment of glaucoma

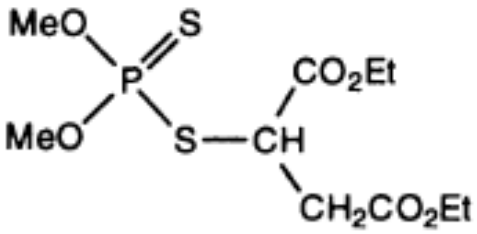


Insecticides

- Parathion and malathion are non toxic towards humans
  - The phosphorus/sulfur double bond prevents these molecules from antagonizing the active site on cholinesterase enzyme
  - The equivalents compounds containing a phosphorus/ oxygen double bond are lethal compounds



PARATHION

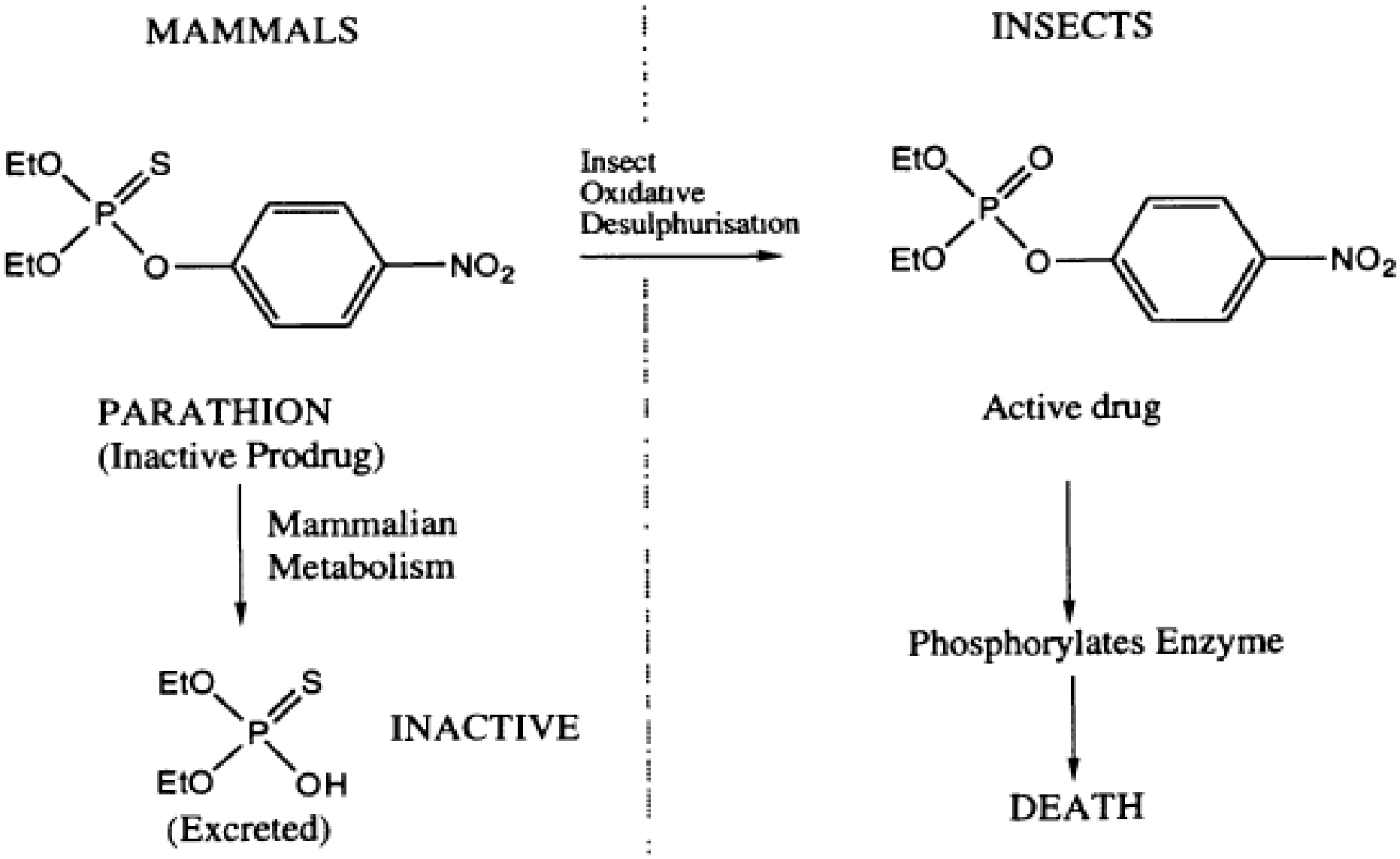


MALATHION

Cont…

- Luckily, no metabolic pathways exist in mammals that can convert the phosphorus/sulfur double bond into phosphorous/oxygen double bond
  - But this pathway exists in insects
  - In insects, parathion and malathion are Prodrugs
  - Oxidative desulfurization leads to active anticholinesterases which irreversibly bind to insect's acetylcholinesterase and lead to death

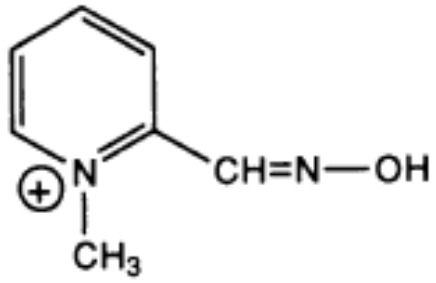
Metabolism of insecticides



Pralidoxime-an organophosphate antidote

- Search for a drug which will displace the organophosphate molecule from serine.
  - This requires hydrolysis of the phospate-serine bond, but this is a strong bond and not easily broken
  - Therefore, a stronger nucleophile than water is required

- Hydroxylamine can hydrolyze such a covalent bond



Hydroxylamine group reaction with phosphate ester

